31. Write a short note on each of the following :

 (i) Use of 'Peer assessment' in evaluating student learning. (B) [Ans. 2:04:3:01]

 (ii) Use of self-assessment in evaluating learning (B) [Ans. 2:04:3:02]

32. Explain the methods of 'peer assessment' and 'self-assessment' in evaluating student learning. (A) [Ans. 2:04:3:01 + 2:04:3:02]

33. State the uses of peer and self-assessments in evaluating student learning. (B) [Ans. 2:04:3:03]

# UNIT – III

# TOOLS AND TECHNIQUES FOR CLASSROOM ASSESSMENT

## 3:00 Introduction

In this unit, important tools and techniques for classroom assessment like observation, self-reporting techniques, anecdotal record, check list, rating scale, different kinds of tests (like achievement test, diagnostic test, prognostic test etc.), rubrics, its purpose and importance, steps to create a rubrics by students, assessment tools for affective domain like attitude scale, motivation scale and interest inventory, different kinds of test items and principles of constructing test items, Major issues in the assessment of learning, particularly commercialization of assessment, poor test quality, domain dependency, measurement issues and system issues, Reforms undertaken in the assessment of student learning including 'Open book system' and 'Online Examination', Examination reform reports presented by 'A.L. Mudaliar Commission', 'Kothari Education Commission' and 'National Policy on Education' are to be discussed.

## 3:01 Tools and Techniques for Classroom Instruction

In order to assess the learning achievement of students and their method of learning, it requires for the teacher to collect various relevant information related to them. Depending upon the purpose of assessment and the nature of information to be collected, classroom teacher has to use the appropriate tools and techniques of assessment.

Methods of organising the various activities that find place in an assessment process are called 'Assessment Techniques' and the devices and materials employed in them are known as 'Tools of assessment'. For example, if 'observation' is used in the collection of assessment information, the observational activity is the 'technique' and the devices used in it like 'checklist', 'rating scale' etc. are the 'tools' of assessment. Further, in observation there are four different kinds of techniques - (i) Participant observation, (ii) Non-participant observation (iii) Uncontrolled observation and (iv) Controlled observation. Among these techniques, the most appropriate one is to be selected and used, according to the purpose of assessment, nature of the information to be collected and the situation in which it is collected.



Generally, among the tools and techniques employed in classroom assessment, the following are the important ones :

i) Observational techniques

ii) Self-reporting techniques

iii) Anecdotal Record

iv) Check List

v) Rating Scales

vi) Different kinds of tests

## 3:02 Observation

Observation is nothing but keenly watching the external behaviour of persons in appropriate situations, controlled or uncontrolled. It is concerned, with neither what a respondent places on paper, nor with what he says in an interview **(C.V. Good).**

According to **Young**, observation is careful and systematic viewing of a selected situation and recording then and there, what is perceived.

In recent times observation techniques have been refined and made more objective and reliable. Wherever necessary, audio and video recording devices, telescopes and measuring instruments like thermometers, audio-meter, stop-watch etc. are used. However, observation is considered as a non-testing device.

**An Example for observation**

A teacher can easily guess that a student is highly anxious and excited, by observing external symptoms like trembling of the hands, incoherent speech, biting nails, restlessness and similar such behaviours.

## 3:02:1 Procedures to be Followed for Good Observation

i) In appropriate situations, observe the whole event.

ii) Observe only one aspect of an individual's behaviour at a time.

iii) Observe, without the knowledge of the observed and record then and there what is observed.

iv) Observer should not mix his opinions and guesses with the observed data.

v) Observation should be continuously carried out, within the time schedule.

## 3:02:2 Steps Involved in Carrying Out a Good Observation

i) Proper Planning



a) Determining the details to be collected through observation: The behaviour aspect / incident to be observed, should be determined before hand.

b) The individual or group to be observed should be precisely indicated.

c) Observers should be given clear briefing about places where observation should take place, timings, time interval, the minimum number of times an incident / aspect should be observed and similar other predetermined details.

d) Selecting the method of recording the observations then and there and also the tools to be used (Check List, Descriptive Rating Scale, Score Cards, Forms to record tally marks etc.)

ii) Executing the observation skillfully and recording the data carefully.

iii) Studying and interpreting the recorded data.

## 3:02:3 Types of Observation

There are several types of observations such as participant observation / Non-participant observation, Natural / artificial observation and Scheduled / unscheduled observation.

### 3:02:3:01 Participant Observation

In participant observation, the observer will find a place in the group where the observed is a member. By participating in the group activities, he can know very well the situation in which the group functions. The observer can get the factual information about how the observed person behaves in the group, his performance, attitudes, sociability, leadership and similar other personality traits.

### 3:02:3:02 Non-Participant Observation

Non–participant observation is best suited to observe infants, maladjusted individuals and similar other persons. In this, the observer takes a position at a place where his presence goes unnoticed and is also least disturbing to the group, from where he can observe in detail the behaviour of an individual under observation or some specific characteristics of a group. It permits the use of recording instruments and the gathering of large quantity of data.

Of the several observation methods, observing in natural situation (uncontrolled observation) and artificial or controlled observation are the only two types, largely used in education. We shall discuss them briefly.



### 3:02:3:03 Uncontrolled Observation

Observing students in the classrooms, play–ground, library and common places without their knowledge is an example of uncontrolled observation.

### 3:02:3:04 Controlled Observation

We may ask a child to study in a silent room and then in a room with a lot of noise with a view to compare its learning achievements in these two different situations. It could be found that the child's learning efficiency is more in a silent atmosphere and less while it studied in a noisy environment producing distractions. Moreover, when the child studies in a noisy atmosphere, it has to take more efforts and as a result gets tired quickly. In this experiment, noise level is controlled and the child's behaviour (learning outcomes) is compared in opposite situations. This experiment is an example for controlled observation.

## 3:02:4 Uses of Observation

By observing a person, we are able to understand his mental state like happiness, sorrow, anger, excitement and other similar moods. In colleges of education, we try to evaluate a student - teacher's performance or teaching ability through observation. To know about the personality of a person, observation tools are useful. Observing a person's behaviour scientifically, carefully and intelligently is required to understand his personality clearly. The other advantages of observation are :

i) Observation is suited not only for individuals but also for groups.

ii) Observers need only a short-term training for effective functioning.

iii) This method suits all age groups and both the genders.

iv) This method is not expensive and may need only a few gadgets.

v) Observing an individual or a group in the natural environment provides reliable information.

vi) Observation tools can be designed to meet any situation.

### 3:02:5 Limitations of Observation as a Technique of Assessment

i) The personal likes and dislikes of the observer and his own limitations will affect the quality of observation.

ii) Only expressed behaviour could be observed; the inner feelings cannot be found out.



iii) Recording may not be done on the spot; the data recorded may not be accurate.

iv) Observation requires more time, more patience and a keen insight.

## 3:03 Self-reporting Techniques

In this kind of techniques the respondent himself provides answers to items in the given questionnaire, concerning his characteristics or behaviour, either orally or in the written form. The investigator gathers answers from the respondents by reaching out the questionnaire to them either in person, through post or phone. In interview too, the interviewer asks questions to the interviewee and gets his answers directly in a face to face situation, besides noticing his mood, likes, dislikes, fears etc. Generally for assessing attitudes, interests, adjustment in behaviour and other personality traits, self reporting techniques are extensively used.

### 3:03:1 Examples for Self-reporting Techniques

The following are the important self-reporting techniques used in the assessment of students.

i) Questionnaire

ii) Opinionnnaire

iii) Check list

iv) Interest Inventory

v) Attitude Scale

**Woodworth's Personal Data Sheet, Bell's Adjustment Inventory, The Minnesotta Multiphasic Personality Inventory (MMPI), Thurstone's Occupational Interest Inventory, S.P. Ahluwalia's Teaching Attitude Inventory, Strong-Campbell Interest Inventory,** are some of the popular self-reporting devices used throughout the world.

## 3:03:2 Advantages of Self-reporting Techniques

1. As answers are obtained directly from the respondents, the gathered responses / information are highly valid.
2. Large amount of data can be obtained very quickly and cheaply.
3. Left out data could be easily obtained back; hence the data collected using self-reporting techniques are highly reliable.
4. Data obtained from closed form of questions can be easily tabulated and analysed.



## 3:03:3 Disadvantages of Self-reporting Techniques

i) We cannot expect that all the respondents will be giving answers or expressing their ideas to items in the self-reporting device completely and honestly without any hiding.

ii) Generally, respondents attempt to give answers to items that are acceptable to all. Therefore it becomes necessary to check the truthfulness of the collected data by cross checking them through other sources i.e. the reliability of the data collected using self-reporting devices is mostly low.

iii) There are chances for questions being misunderstood and answered accordingly.

iv) The answers provided by the respondents may be influenced by their mood and emotions at the time of answering the self-reporting device.

v) Most of the self-reporting devices contain fixed choice questions and lack flexibility. No opportunity is given to include supplementary or explanatory information.

vi) Low response rate.

## 3:04 Anecdotal Record

### 3:04:1 Meaning of Anecdotal Record

This is associated with observation. In everyone's life, particularly while young, a few note worthy incidents / events occur. These incidents picturize the nature of his mind and his abilities. Recording such episodes and compiling them is called anecdotal record. Parents will note down the episodes in the life of their children. But most of them will be subjective. They may distort the truth atleast to some extent.

Teachers should keenly observe their students and record the significant events or incidents that happen in their school lives, then and there in their respective records. Thus anecdotal record is **"a factual description of the meaningful incidents and events which the teacher has observed".** As the anecdotal record highlights the student's personality traits, needs, problems in social adjustment, special abilities and such other information, it is useful to a great extent in providing guidance and counselling to the concerned student.



### 3:04:2 Definition of Anecdotal Record

Anecdotal Record is the computing of significant incidents in the life of a student. It is a compilation of factual descriptions of meaningful incidents and activities relating to students which were keenly observed by the teacher.

**Raths Louis** says, "Anecdotal record is a report of the informations about the significant episodes which happened in a student's life."

According to **Traxler,** "Anecdotal record as the name implies involves setting down accounts concerning some aspects of pupil behaviour which seems significant to the observer."

An Anecdote is an isolated incident of the behaviour of the student which has some special significance. Anecdote can be compared to a snap shot of a camera which catches, the pose of an individual at a specific time among the sequel of actions.

### 3:04:3 Salient Features of Anecdotal Record

i) Factual information about the important events in the student's life will be recorded objectively. Insignificant daily routines of day to day life will not find a place in the record.

ii) Only one incident at a time will be observed and recorded.

iii) Only one incident will be recorded in a card. Each card will be of 3" x 5" size.

iv) Cards with entries of events will be arranged chronologically and maintained as a record.

v) Every teacher will maintain separate cards.

vi) Behaviours from incidents happening in various places like play ground, hostel, laboratory, excursion / field trip, assembly hall, rest room and other places will be collected and included in the anecdotal record.

vii) Description about an incident / event and the teacher's comments / interpretations on it should be kept separately.

viii) Generally, anecdotal records will be kept confidentially; notings about succeeding events will be added cumulatively.

ix) It will not be considered as a reliable strategy to collect information about the students; it is only supplementary to other methods of data collection.

### 3:04:4 Maintaining Anecdotal Record

Teachers should decide about which students have to be observed, when to undertake observation, and how to record the information obtained.



Everyday a specific time should be earmarked to record the observations made. Some teachers would like to record the observations then and there; some others fix separate timings for observation and recording. If there is a long interval between observation and recording, the teacher may forget some of the details observed; he may add some observations which were never made. These affect the objectivity of the recorded notings.

Instead of thinking of collecting anecdotes regarding problem students only, recording of numerous anecdotes of all students is far better. With the continuous supervision and help of the school counsellor, teachers should keep maintaining anecdotal records. At the end of every year the informations recorded in the anecdotal record will become part of the cumulative record and become an information file for providing future guidance.

Anecdotal incidents may be recorded in small sheets of papers or recorded in the blank forms provided by the educational institutions and then integrating them into small cards. Two model anecdotal forms are given in the following page :

**Anecdotal Record Form I**

| Name of the Observed Student | Observer | Date |
|---|---|---|
| Std : | Place where observation took place | |
| Description about anecdotal incident<br><br><br><br>Signature of the Observer | | |

**Anecdotal Record Form II**

| Name of the Student Observed | Observer | Date |
|---|---|---|
| Description of the incident in brief<br><br>Comments :<br><br>Recommendation :<br><br>Signature of the Observer | | |



### 3:04:5 Advantages of Using Anecdotal Record

1. Student's behaviour is observed under varied situations, and as only reliable information is collected, this record is helpful in properly understanding the student's personality.
2. It is useful to cross check the information obtained from other sources.
3. As the teacher is interested in his students and try to observe them continuously, he gets opportunities to know about his students fully and get closer to them.
4. Collecting information using the anecdotal strategy is very useful in studying the behaviour of small children and the mentally retarded.
5. The teacher, instead of criticizing his students generally, it will be more productive to offer his ideas and recommendations on the basis of accurately observed behaviour.

### 3:04:6 Limitations of Anecdotal Record

i) As activities like observing anecdotal incidents, recording them and studying them take a lot of time, the work load of teachers increase.

ii) When the present behaviour of the student differs very much from the recorded behaviour, other teachers who by chance look into the anecdotal record may develop a wrong notion about the concerned student.

iii) As the incidents recorded in the anecdotal record happens to be the important events in the student's life, any exaggerated description may lead to misunderstand the student.

iv) Informations recorded in the anecdotal record will be useful, only if they are accurate and comprehensive.

v) If the described behaviour is isolated from the social background in which it happened, its real nature cannot become clear.

vi) The chances of occurrence of such incidents which reveal the impcrtant behavioural traits of students and that too in adequate numbers taking place now and then, is very much limited.

## 3:05 Check List

It is a simple laundry-list type of device, consisting of prepared list of items. It is used to record the presence or absence of the item by checking 'yes' or 'no' or, the type or number of items present may be indicated by inserting the appropriate word or number. (Responses to the check-list items are a matter of fact and not judgement or opinion)



Check-list has the advantage of recording the systematized observations covering almost all the important aspects of the object or act observed. Thus check-list is extensively used in educational surveys to get facts and also in observational studies of behaviour. When used as a sort of scale to yield a score, it is an instrument often used in educational appraisal studies of school buildings, text books, facilities available (recreation, laboratory, library etc.).

(E.g.) Put a tick (✓) mark in the bracket against the following statements which you consider as describing you correctfully.

i) I make friends easily ( )

ii) I like to go alone in long distance walking ( )

iii) I take much care about my belongings ( )

iv) Generally, I think a lot about myself ( )

### 3:05:1 Uses of Check-list

Check list is useful in collecting educational statistics, to find out the needs of students, facilities available in educational institutions and to observe and record students behavioural traits. They are also useful to find out the student's learning efficiency and evaluating the effectiveness of classroom teaching-learning activities.

### 3:05:2 Merits of Check-list

i) The checklist could be used easily by the investigator or the informants/respondents

ii) It takes only a few minutes to fill in the details to be provided as they are indicated using tick mark (✓) or words or numbers.

iii) As informations are obtained through observation, there is no need to doubt them.

iv) Each segment of the information or item could be used easily to compute its frequency, average and percentage.

### 3:05:3 Limitations of Check-list

i) Checklist is not helpful to gather complex information.

ii) The objectivity of the check list cannot be stated with certainty.

## 3:06 Rating Scale

Check list is useful to observe a thing, event or behaviour like performance of tasks, levels of skills exhibited, procedures followed, processes involved, human behaviours expressed etc. to find whether a certain characteristic trait or attribute is present or not. But rating scale is superior to check list, as it not only



finds out whether a certain characteristic trait or attribute is present or not but also evaluates how much of it is present.

Stated briefly **rating scale is a device containing a set of graded categories in small numbers that helps an individual to express his / her judgement regarding how much certain characteristic traits or attributes are present in some situation, object or behaviour.**

### 3:06:1 Definition of a Rating Scale

**Ruthstrong** defines a rating scale as "essentially a device, for direct observation. It is a device having a set of few graded categories to judge whether a given characteristic trait or attribute is present or not in an observed object or a situation and if present, how much of it".

According to **A.S. Barr** and others, "Rating is a term applied to expression of opinion or judgement regarding some situation, object or character. Opinions are usually expressed on a scale of values. Rating techniques are devices by which such judgements may be quantified".

### 3:06:2 Constructing a Rating Scale

In a 'Rating Scale' only a small number of items relating to the characteristics or attributes of a thing, event or behaviour will be given. On the basis of the findings of the observation, each item will be given a rating by the observer to indicate the degree to which each attribute is present. Such rating could be done in the following three ways.

### 3:06:2:01 Numerical Rating Scale

In this method, to indicate the rating, serial numbers are used like 1,2,3,4,5 or -2, -1, 0, 1, 2 or symbols like A,B,C,D,E etc. Generally 1 is considered to be the lowest and 5 is the highest; similarly A is considered to be the lowest and E as highest. Usually Rating Scales have 3,5 or 7 points.

(e.g.)

Numbers : 1. Very Poor 2. Poor
3. Average 4. Good 5. Very Good

(Instruction : Circle the number you want to indicate for the following statement)

How is 'X' cooperating with fellow students in group work?

### 3:06:2:02 Descriptive Rating Scale

In this method, to mark the rated value, observers will be asked to indicate anyone of the descriptions, given in the verbal form



How is the managerial skill of the teacher?

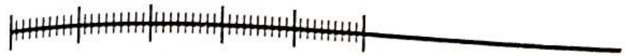
very high High Average Low Very Low

### 3:06:2:03 Graphic Rating Scale

In this method, the rated value is indicated as a dot in a graduated line.

(e.g.) How is the perseverance of 'X'?

(Mark your evaluation as a dot in the following scale given)



### 3:06:3 Uses of Rating Scales

Rating Scales are used in appraisals of various sorts.

i) **Teacher rating :** Selection for the post, for giving promotion, to evaluate efficiency, to sanction increment.

ii) Personality rating for various purposes.

iii) Testing the validity of many objective instruments like paper-pencil inventories of personality.

iv) School appraisals including appraisal of course practices and programmes.

v) Judging the competitors for their proficiency in various competitions (e.g.) elocution competition, essay competition, painting competition, music competition and other similar activities.

## 3:06:4 Merits and Limitations of Rating Scale

### 3:06:4:01 Merits

i) It is easy to evaluate through rating

ii) It is less time consuming

iii) Not much training is needed for the raters.

iv) Various factors like personality, teaching, evaluating the performance of educational institutions, cultural programmes and similar other activities could be assessed separately and later their combined value may be worked out.

### 3:06:4:02 Limitations

i) The rater may find it difficult to understand the qualities to be assessed.

ii) There is the inherent danger of personal likes and dislikes influencing the rating.

iii) In rating, the undermentioned five types of errors may occur.



a) **Generosity Error :** Rating liberally about persons who are very popular or whom one admires.

b) **Constant Severity Error :** Some raters may think everyone else is inferior to them and award low marks only. This kind of error in valuation is called 'Constant severity error'

c) **Average Error :** Some evaluators, by providing average rating to all, would like to escape from possible blame or criticism. This is called 'average error'.

d) **Halo effect :** An evaluator, who has to evaluate various aspects of a person's qualities may be influenced by the high level of the first few qualities and will consider that they must be very good in other qualities also and rate them high. On the other hand if the rating for the first few observed items are poor, they will erroneously award only low marks for all other items also. The reason for this tendency is the superficial general impression is taken as a basis for rating. This is called 'Halo effect'.

e) **Logical Error :** When an evaluator rates closely related or unrelated aspects of qualities, may think logically and allow the influence of one aspect of the quality to enter into another one. For example, while evaluating an answer script, the answers written in beautiful handwriting and without scoring out or corrections may be given high marks by the teacher. There is no relationship between the nature of handwriting and the accuracy of answers. Even then, one influences the other. Similarly though there is close relationship between intelligence and academic achievement, it cannot be said that all intelligent persons are high academic achievers. Logical error is said to occur if the evaluator considers that all high achievers are highly intelligent and rates high accordingly.

## 3:07 Types of Tests

### 3:07:1 Concept of 'Test' in Education

'Test' in education refers to a device containing a set of sequentially arranged questions or tasks to be given to students, requiring them to provide their responses individually. By awarding marks to the responses provided for the test items, students' achievement can be evaluated and compared.

Tests in education hence could be considered as a device that yields measurements by evaluating student-learning as it develops. Test could be conducted daily, weekly, monthly, or during the period of training or course of study. Tests may be in the oral or written form.



### 3:07:2 Categorization of Tests

Different kinds of tests are used, depending upon the purpose. Tests used in education can generally be classified into the following three categories.

i) Achievement Tests : Meant for measuring the achievements of students.

ii) Diagnostic Tests : For finding the deficiencies or shortcomings in student learning.

iii) Prognostic Tests : Measure the capacity of a person for predicting his/her success in a particular field.

Let us discuss about these in detail.

### 3:07:2:01 Achievement Tests

#### A) Meaning and Definition of an Achievement Test

A test that measures the level of performance or the proficiency achieved by an individual after a period of instruction or training is called an achievement test. The marks obtained by a student in an achievement test conducted at the end of a content unit, reveals his level of proficiency in that unit.

**Dowine** defines achievement test as **"any test that measures the attainments of an individual after**

**a period of training or learning is called an achievement test".**

According to **Donald Super,** "Achievement test is a tool to measure the level of proficiency achieved by a pupil, indicating what and how much has been learned or how well a task has been performed".

In an achievement test, one's knowledge, understanding and skills acquired are measured as learning outcomes. It is necessary for the teacher to know how far students have attained proficiency in the subject taught in the class. Unit test, monthly test, terminal examination, annual examination etc. are all examples for achievement tests.

## B) Purposes of Achievement Tests

The purposes or functions of achievement tests are the following :

1. They help to measure the proficiency achieved in knowledge, understanding and skills achieved by students in a specified subject and to determine the relative position of each student in the class.
2. They help in grading students according to their learning achievement.
3. They help in deciding who are all to be promoted to the next higher class / grade.



4. They help to know the entry level knowledge and skills of students at the beginning of an academic year, in different school subjects.
5. Motivating students before providing a new assignment.
6. To test whether the instructional objectives have been achieved or not.
7. To help the teacher to know how far the teaching method employed has been successful.
8. To help the teacher know the fruits of his labour.
9. To inform the parents, the progress of their children in different school subjects.
10. To identify the content areas in which students struggle to learn and arrange for remedial measures.

## C) Types of Achievement Tests

Generally speaking, achievement tests are of two types - (i) Teacher-made and standardized tests and (ii) Criterion referenced and norm referenced tests. Usually they are in the form of written tests.

### i) Teacher-made Tests and Standardized Tests

Teacher-made tests are specific to the syllabus, portions covered, standard in which the students are studying and their general ability. So, other than public examinations, achievement tests conducted in educational institutions or classrooms will vary in content / teaching-units, type of questions, choices given in answering questions and time allotted for completing the test. Through these achievement tests, a teacher can know the learning proficiency of students in his class. But as we are not sure about the reliability and validity of these tests, they are called non-standardized tests.

In standardized tests, subject contents are selected, and based upon which the question paper is prepared, the test items in the question paper are verified by experts, uniform test procedures and scoring system are developed and norms are stated explicitly; they have the essential attributes of good test like validity, relia-bility and objectivity. Public examinations conducted on the state-level / zonal level based on the common syllabus are of standardized test category. Public examinations conducted by State Board of Secondary Education, State Board of Higher Secondary Education, I.I.T. Entrance Tests, National Eligibility Cum Entrance Test (NEET) for admission in Medical Colleges are examples of Standardized tests.



### ii) Criterion referenced and Norm referenced Tests

Tests that evaluate students how far they have achieved proficiency in the subject content and skills in a particular content area are called **'Criterion - referenced tests'**. For example, a test developed to evaluate the learning proficiency of students in the four skills of mathematics i.e. addition, subtraction, multiplication and division will attempt to find the percentage of learning achievement of students in each of these four skills. If a minimum of 80% of marks obtained by students in each of these skills is taken as the acceptable level of learning proficiency then those who score less than 80% marks in each of these skills will be labelled as 'fail'.

Stated briefly, criterion-referenced test, attempts to evaluate the level of learning achievement of students in terms of the objective standard or acceptable level of proficiency. In other words, **criterion referenced tests reveal what individuals can do and that too at what level,**

**Norm-referenced tests** compare the performance score of an individual with the average performance of his group called **norm group,** to determine his/her rank position i.e. students are graded based on the marks they obtained.

### 3:07:2:02 Diagnostic Tests

#### A) Meaning of 'Diagnostic Test'

"A test that helps the teacher to identify the concepts in the content area taught, which students struggle to learn and the nature of the difficulties experienced by them is called a '**Diagnostic Test**'.

Besides this, those tests used to identify the deficiencies of children with learning difficulty, in skills like pronouncing words, reading with comprehension, writing, numerical computation etc. are also known as 'Diagnostic tests'.

#### B) Purposes of Diagnostic Tests

Diagnostic tests are conducted to find the concepts or learning points in a subject, that low academic achievers find it difficult to learn and the remedial measures required to be taken.

**Rose** and **Stanely** (1956) have identified the five levels of diagnosis. They are :

i) In the content area taught, where do pupils have trouble?

ii) Where more number of errors are committed?

iii) Why do such errors occur?

iv) What remedial measures are suggested?



v) How can the errors be prevented?

Among these, the first four are related to the '**Corrective Diagnosis**' of learning difficulties and the fifth one is related to '**Preventive Diagnosis**'.

Briefly stated, the major purposes of diagnostic tests are to identify the concepts in the subject content which students find it difficult to learn, nature of the learning difficulties and the remedial measures required for them.

#### C) Uses of Diagnostic Tests

Major uses of diagnostic tests are the following :

i) Identifying the learning concepts / points that students struggle to learn.

ii) Finding the errors, students commit and their nature.

iii) Identifying the inadequacy in specific skills that are required to understand those concepts / points which students struggle to learn.

iv) For determining which content areas require individualized instruction.

v) Serving as the basis for the modifications required in the teaching method.

vi) Providing feedback to the teacher and students to know their respective shortcomings in the teaching - learning process.

## D) Differences between Achievement Test and Diagnostic Test

| S.No. | Achievement Test | Diagnostic Test |
|---|---|---|
| 1. | Measures the level of performance of students or the proficiency attained in learning in a specific subject or content areas. | Attempts mainly to find the content areas / concepts which the students find difficult to learn. |
| 2. | As different content areas and skills are tested, the range of subject-content tested is wide. | As the content areas / skills which students find difficult to learn alone are tested, the range of content tested is less. |
| 3. | As questions are asked from different content areas, the range of difficulty level of items is wide from easy to very difficult. | In a diagnostic test, to test each concept or learning skill, many items are formed and clustered as a group called sub-test. Through such sub-tests only, students' learning difficulties are identified. All the test items are simple and easy. |
| 4. | Time for completing the test is well defined. | No time-limit set for completing the test. |
| 5. | Test items are not arranged according to the sequence of the content areas found in the syllabus. | As the concepts / learning points in every lesson are arranged in order, test items developed around them get sequentially arranged as per the order of the content areas in the syllabus. |
| 6. | Students' responses are scored to measure the proficiency in learning. | Only errors committed by students are taken into account so as to assess the nature and level of difficulty faced in learning the particular content unit / concept. |
| 7. | Used to grade pupils or promote them to next grade / class. | Following diagnostic test, remedial teaching will be undertaken. |
| 8. | Comes under 'Summative Evaluation'. | Comes under 'Formative Evaluation' |

## 3:07:2:03 Prognostic Test

### A) Meaning of a Prognostic Test

A Prognostic test is one which measures the capacity of a person for success in some field and as such it is predictive in character. **In education prognostic test is nothing but an aptitude test.** Aptitude tests help to predict the potential abilities and interests in a person for a particular vocation or course of study.

According to **Hull**, "aptitude test is one that measures the inborn capacity to learn a specific vocation or skill." A number of aptitude tests are in vogue to test different kinds of aptitudes. Present day psychologists use aptitude tests like musical aptitude test, mechanical aptitude test, clerical aptitude test, teaching aptitude test, scholastic aptitude tests for different courses of study etc. As aptitude test attempts to predict the natural capacity of a person in a specific field, this kind of prognostic test is conducted before start imparting training / instruction in that field. As aptitude tests predict before hand, how much a person will get success in a particular field on getting training, they are also known as 'Pragmatic Tests'.

## B) Some Well-known Aptitude Tests

### a) Seashore's Musical Aptitude Test

This has been developed by analyzing the musical skills. Music aptitude Test tries to test the knowledge regarding pitch and intensity of sound, sense of tonal discrimination and tonal memory, ability to identify sound beats and notations, etc.

### b) Mechanical Aptitude Tests

These could be either verbal tests or tests requiring the testees to put together various parts of mechanical devices. Guessing the full shape that will emerge if the given components are assembled together, judging which tools are appropriate for what type of activities, speed of action, knowledge about the details of machines and similar other informations will be tested in written tests. In the tests for assembling things, items like parts of cycle, bell, lock, screwing type pen etc. will find a place. To test spatial perception, 9 curved wooden blocks making up a big wooden block (Wiggly Blocks), a block cut into 27 small cubes etc. will be used.

### c) Differential Aptitude Test (DAT)

This integrated battery of tests was developed by **George K. Bennett, Harold G. Seashore** and **Alexander G. Wesman** in 1947 and revised in 1963.



There are 8 sub-tests in it, which are described as under:

*i)* ***Verbal reasoning :*** It has the items of the double analogy type.

*ii)* ***Numerical ability :*** It consists of numerical problems, emphasizing comprehension rather than simple computation facility.

*iii)* ***Abstract reasoning :*** It consists of a series of problem figures.

*iv)* ***Spatial relations :*** In it a diagram of flat figure is shown. The examinee has to visualize and indicate which solid figure could be produced by folding the flat figure.

*v)* ***Mechanical reasoning :*** The examinee is shown a diagram of a mechanical device or a situation is shown and the examinee must indicate which choice is true to the situation.

*vi)* ***Clerical speed and accuracy :*** Each item is made up of a number of combinations of symbols one of which is underlined. The examinee must mark the same combination on his answer sheet.

*vii)* ***Language use-spelling :*** A list of words is given, some of which are mis-spelt. The examinee has to indicate the items which are incorrectly spelt. [(e.g.)

Proffessor. (This word is written with wrong spelling. Two 'f's will not come in this word).

viii) ***Language use-sentences :*** A sentence is given, divided by marks into 4 sub-sections. The examinee has to indicate which section contains errors.

(e.g.)

1. Discussing with his father what to do next
A B
the train left the station leaving him behind
C D

Here the mistake is committed in part A. It should be "While he was discussing with his father". What is given as such, means that the train was discussing with his father.

2. The Auditor said that he had only found
A B
two errors in the accounts
C

Here the error is in 'B'. (the correct usage is "that he had found only")

## C) Uses of Aptitude Tests

i) Selecting students and teachers for a course of study or vocation.

ii) For providing educational and vocational guidance for pupils.



iii) Selection and training of people, based upon aptitude tests, will be socially productive in the long run and will also yield maximum job satisfaction for the incumbents.

## D) Differences between Aptitude and Achievement Tests

i) Achievement tests are conducted after a period of training or instruction in a field whereas aptitude tests are conducted without providing any such training or instruction.

ii) Achievement tests furnish the standard or criterion against which the predictive effectiveness of aptitude tests can be judged.

iii) Aptitude and achievement tests differ in their purposes and not often in content. Aptitude tests measure present capacity to predict future achievement whereas achievement tests are interested in the present level of performance only.

iv) Aptitude tests unlike achievement tests, take into account the direction and intensity of the individual's interests also.

## 3:07:2:04 Other Categories of Tests

### A) Ability Tests

These tests try to assess the general ability of an individual. Ability tests meant for those with a particular level of educational qualifications (like matriculates, graduates, post-graduates etc.) do not consider the syllabus of any specific subject or course unlike an achievement test. Ability tests try to assess the general abilities expected of persons with educational qualifications at a particular level of education, at a given point of time. National level selection or admission tests for candidates hailing from different regions / universities are good examples for an ability test.

### B) Oral Tests

In this kind of tests, the teacher asks questions and get responses from students in the form of oral replies or performance.

Let us see the details of these two types of oral tests.

#### 1. Oral Response Test

For assessing the content knowledge developed in students as well as to attract their attention on what are taught in the class when teaching is in progress,



the teacher ask questions and get students' responses in the form of oral replies. For assessing language skills, **oral response test** is highly useful. Further to find whether students are following the demonstration of an experiment carried out in the classroom and understanding the procedure and precautions, the teacher can ask questions and get oral replies from students. This also promotes training in observation in students. At the pre-primary stage of education, learning achievement is assessed mostly through oral tests. Oral tests are economical as compared to written tests. When students are taken out for excursion or to a movie related to education, to test the learning outcomes of students, oral tests could be used.

#### 2. Oral Performance Tests

Generally in performance tests, testees are assessed how efficiently they complete the given task within the stipulated time. In oral performance tests, students are required to execute the task given to them in the form of verbal acts. Recitation of poems in the language class, recitation of a multiplication table in the class, speaking on a given topic, participation in a debate, presenting the project report and explaining it, participation in oral quiz competition etc. are examples for oral performance test.

## 3. Practical Tests

In the teaching - learning process, particularly in science subjects and vocational courses, student's practical work is important. In professional courses like medicine and engineering too practical examinations are conducted to assess the skill development in students.

Generally, practical examinations aim to assess the proficiency of students' skills related to the laboratory work in carrying out experiments efficiently and neatly, without wasting time. Practical examinations evaluate students' ability to handle apparatus and equipments, observation skill to identify and separate the parts of thing properly, reading and recording the measurements accurately etc. However, it is difficult to implement practical tests at the high school education level, as it costs more money and time. Further, it is also difficult to select experiments for students' practical work, grade / class wise.

In teacher education courses, practical examinations are more important. By evaluating proficiency in the various teaching skills like motivating, explaining, using the technique of stimulus variation efficiently, using the blackboard, using reinforcers, asking questions, closure of teaching a topic, conducting classroom demonstration etc. teacher trainees could be graded for their teaching competency.

Stated in a nutshell, putting in practice of those that are learned is called 'practical work'. Evaluating the efficiency or proficiency in executing this, is the aim of practical examinations.

## 3:08 Rubrics

The term 'Rubrics' denotes a guide listing a coherent set of rules or specific criteria for scoring or grading students' answer papers, assignments, projects and academic performances.

In instructional settings, rubrics clearly define the academic expectations or the learning standards for students and help to ensure consistency in the evaluation of academic work from student to student, assignment to assignment or course work to course work. Rubrics are also used as scoring instruments to deter-mine grades or the degree to which learning standards have been demonstrated or attained by students.

Stated briefly, **rubrics denotes the criteria used by the teacher to assess students' learning achievement and their specifications describing**

**the different levels of performance.** For example, when a teacher is grading a particular assignment of students on a scale 1 to 4, the rubrics may detail what students need to do or demonstrate to earn a point, 1,2,3 or 4 i.e. details like what components if present in the assignments will fetch the grade 1, along with these if what else are present will fetch the grade 2, and what are all to be present in the assignment to fetch the highest grade of 4 etc. will be indicated in the rubrics.

### 3:08:1 Purpose of Rubrics

Like any other evaluation tool, rubrics are useful for certain purposes and not for others. The main purpose of rubrics is to ensure consistent and objective evaluation of student performances. For some performances, the teacher has to observe the student in the process of doing something, like using an electric driller or discussing an issue. For other performances, the teacher should assess the product resulting from the student's work, like a written report or a chart or learning aid prepared. The following table shows some of performances that can be assessed with rubrics.



| Types of Performance | Examples |
|---|---|
| ***Processes***<br>✦ Physical skills<br>✦ Use of equipment<br>✦ Oral communication<br>✦ Work habits | ✦ Playing a musical instrument<br>✦ Preparing a slide for the microscope<br>✦ Making a speech to the class<br>✦ Reading aloud<br>✦ Working independently |
| ***Products***<br>✦ Constructed objects<br>✦ Written essays, reports, term papers<br>✦ Other academic products that demonstrate understanding of concepts | ✦ Wooden bookshelf<br>✦ Handmade apron<br>✦ Water colour paintings<br>✦ An assignment report on the drama traditions followed during the period of Shakespeare<br>✦ A research report on the influence of Marshell's theory on economy<br>✦ Model of Diagram (of structure of an atom, structure of flower, planetary system etc.)<br>✦ Concept Map |

### 3:08:2 Example of Marking Rubrics

| Marking Rubrics | Excellent | Good | Average | Poor |
|---|---|---|---|---|
| Compre-hension | Demonstrated complete knowledge of concepts and principles in the content area. | Reflected most of the knowledge or main points of concept or principles. | Showed practi cal knowledge of some points of the concept in the content area. | Showed minimal knowledge of concepts in the content area. |

| Marking Rubrics | Excellent | Good | Average | Poor |
|---|---|---|---|---|
| Compre-hension | Showed an excellent under-standing in interpreting the concept of the content | Showed good understanding in interpreting the concepts in the content. | Showed some understanding of the concepts in the content; it could have been still better. | Poor in interpreting the concepts in the content area. |

### 3:08:3 Steps in Students Learning to Create a Rubric

An example for developing a seven step rubric, for assessing an assignment related to writing skill has been presented below :

i) Teacher explains with illustrations what makes an assignment on a topic, excellent and ask the students to reflect on them and review.

ii) Teacher takes a topic as an example and describes the **elements** or **attributes** of a good assignment on the topic (which are otherwise known as assessment criteria) and get feedback from students.

iii) Based on the feedback from students, the teacher improves the criteria of assessment (i.e. rubrics) he has developed.

iv) Grading the attributes / criteria developed to form the rubrics (i.e. arranging which one is exemplary, the one that is good, average etc.)



v) The teacher using the rubrics he has developed and improved, actually evaluates some student assignments on that topic, in the presence of students and discusses regarding the assessment he has made.

vi) Using the rubrics developed by the teacher, students assess their own assignments on that topic.

vii) Students themselves develop a rubric for a topic and evaluate their own and other student's rubrics.

### 3:08:4 Importance of Rubrics

i) Rubrics helps the teacher to assess student's performance and their work using standardized set of rules i.e. rubrics help the teacher to assess students' performance transparently and consistently, in all kinds of academic work.

ii) To write rubrics, teachers need to focus on the criteria by which learning will be assessed. This focus, on what the teacher intends students to learn rather than what he intend to teach, actually helps to improve instruction. Rubrics help the teacher to focus not only on the content he teaches but also the learning outcomes that result from his teaching.

iii) Rubrics help the students to understand the important concepts in a content area that are to be learned and while they express them in their performance which aspects should get more focus. Keeping them in mind student should get engaged in learning the content area.

iv) Rubrics help students to reflect on their academic work and understand the areas which require more concentration and efforts so as to improve their performance.

v) Rubrics help to assess the learning achievement of students explicitly and fairly, in all fields including fine arts such as music, dance, drama, drawing and painting, as well games, producing handicrafts etc.

vi) Rubrics help to increase transparency and objectivity in assessment.

vii) Rubrics help students to assess their own learning and performance. The performance–level descriptions in rubrics help students understand what the desired performance is and what it looks like.



## 3:09 Assessment Tools for Affective Domain

### 3:09:1 Three Learning Domains

Whatever may be the learning, it takes place in three domains. They are : (i) Cognitive domain (ii) Affective domain and (iii) Psycho-motor domain (also called as conative domain).

Cognitive domain is related with thinking and other mental activities. It is mainly concerned with acquiring knowledge and putting it into practical use. Affective domain is related with emotional development, aiming to cultivating interests, attitudes, values and appreciations in the learner. Psycho-motor domain involves the acquisition of various physical skills which involve neuro-muscular coordination.

### 3:09:2 Affective Domain and the Tools Used to Assess Learning in it

Learning in the affective domain involves desirable changes in interest, attitudes, values and feelings. Learning in the affective domain help the learners to have better adaptability to the demands of the society. As compared to the cognitive domain, learning situations and processes are not so well defined in the affective domain. Moreover, the assessment of effective learning in the affective domain may not be as precise as it can be for cognitive learning.

In the cognitive domain, the main organising principles of learning are from simple to complex, from concrete to abstract. In the affective domain the main organising principle is the degree of **'internalization'** i.e. the extent to which the feelings or emotions are controlled / modified and get incorporated by the learner and reflect it as part of his / her personality. The main stages in affective domain learning are : (a) Receiving (b) Responding (c) Valuing (d) Organisation and (e) Characterization.

**a) Receiving :** This refers to the learner's sensitivity to certain stimulus or pattern of stimuli (Phenomena). Receiving consists of (i) awareness of the stimuli (ii) Willings to receive and (iii) show selected attention.

The above behavioural elements get reflected in the affective behaviour **'appreciate'.**

**b) Responding :** This level goes beyond the receiving level. After perceiving the stimulus, the learner responds to it. This corresponds to expressing one's **'interest'.**

**c) Valuing :** This level implies perceiving a concept or any belief as having worth and consequently revealing consistent preference or commitment in behaviour towards it.



This behaviour is considered as expressing one's **'involvement'.**

**d) Organisation :** For situations where more than one value is relevant, the learner organizes the values into a system and also determines the inter-relationships among them.

This kind of behaviour is considered as expressing one's **'attitude'.**

**e) Characterization :** After the values have been organised in individual's mind, they control his / her behaviour. This is the stage of internalization. At this level, the learner is capable of practising or acting on his/her values and beliefs.

Stated in brief 'affective domain related learning' refers to development of desirable changes in interest, motivation, attitudes and values. Assessment of affective domain related learning involves the assessment of attitude, motivation, and interest. The tools used in the assessment of these are :

i) Attitude Scale

ii) Motivational Scale

iii) Interest Inventories

Let us discuss them in detail.

## 3:09:3 Attitude Scales

According to **Frank Freeman,** 'An attitude is a dispositional readiness to respond to certain situations, persons or objects in a consistent manner which has been learned and has become one's typical mode of response. An attitude has a well-defined object of reference. For example, one's views regarding a class of food or drink (such as fish and liquors), sports, Maths or Democracy are attitudes."

By gathering one's personal views towards a particular object, idea or incident and quantifying it in numbers, his/her attitude could be predicted.

Attitude scale is a tool useful in gathering one's personal views towards a particular object, idea or incident and quantifying it in terms of numbers.

A number of statements expressing various points of view towards the issue in question (it may be a social group, institution, object, incident, idea or practice) are gathered and given to subjects, asking them to mark those statements with which they agree and to what extent. From the information thus collected the attitude of those responded for the statements is assessed.

Generally there are two types of attitude scales to find the attitude of an individual. They are : 1. Thurstone's Equal



Interval Scale of Attitude 2. Likert's Summated Scale of Attitude.

### 3:09:3:01 Thurstone's Equal Interval Scale of Attitude

A number of statements usually 20 or more that express various points of view towards the issue in question (it may be a social group, institution, idea or practice) are gathered. For example to assess the attitude of people towards church, a set of statements like the ones given below are constructed and given to the subjects, asking them to mark those statements with which they agree.

| S.No | Statements | Subject's Response | |
|---|---|---|---|
| | | Yes | No |
| 1. | I like the ceremonies of my church but do not miss them much when I am away | | |
| 2. | I think the church is a parasite on society. | | |
| 3. | The church is needed to develop religion, which has always been concerned with man's deepest feelings and greatest values | | |

In the sample given above, only two extreme attitudes (Items 2 and 3) and a neutral position (Item 1) towards the church are presented. But in the full scale there are items indicating many intermediate positions. Totally there may be 40 or 50 items in an attitude scale. These statements have all been already calibrated by a large number judges on an eleven point scale. This is done by asking the judges to place each statement in one of the eleven files, according to the degree of favourableness or unfavourableness of each item with respect to the question / issue on hand (very highly unfavourable item is to be placed in the first file and very highly favourable item in the eleventh file). The median value of the judged locations for a statement is its **scale value** or **weighted score**.

These statements are given to the respondents whose attitude towards the issue on hand is to be measured, and asked to express their opinion by putting a (✓) mark against each of the statements. For each of the statements with a tick (✓) mark, appropriate weighted score will be given and the sum total of the scores thus obtained by an individual will be taken to be the measure of his / her attitude towards the issue.

## 3:09:3:02 Likert Summated Rating Scale

The most widely used procedure today to assess attitude is the **Likert Method**. In this, statements concerning a question / issue on hand, are not assigned a weighted score with respect to the issue. Instead, the subject indicates the degree of his agreement or disagreement with each statement by



marking a position on a scale. Often the scale has three or five intervals which vary from extremely negative to extremely positive. For example.

Strongly Agree (S.A), Agree (A), Not Sure (N.S), Disagree (D.A), Strongly Disagree (S.D.A).

*Example*

| S No. | Items to measure the attitude towards Professional Education | Fully accep-table | Accep--table | No Opinion | Unaccept-able | Fully unaccep-table |
|---|---|---|---|---|---|---|
| 1. | For admission to professional education courses, only marks secured in the XII Std. Public Examination should be taken into account. | | | | | |

| S No. | Items to measure the attitude towards Professional Education | Fully accep-table | Accep--table | No Opinion | Unaccept-able | Fully unaccep-table |
|---|---|---|---|---|---|---|
| 2. | Entrance tests are essential for admission to professional education courses. | | | | | |
| 3. | Any one with prescribed minimum qualification may be permitted to join a professional course. | | | | | |

For positive statements the responses are scored as under : 5 for (S.A), 4 for (A), 3 for (N.S.), 2 for (D.A) and 1 for (S.D.A). For negative statements the scoring pattern is reversed. The scores thus assigned for the responses given by an individual for the different items in the attitude scale are **summed** up to get the attitudinal score of the individual.

### 3:09:3:03 Uses of Attitude Scales

i) Attitude scales are useful to find out the likes and dislikes of the students.

ii) To understand some of the peculiar behaviours of the students, it is necessary to find out their attitudes as attitudes play a big role in shaping behaviours.

iii) As group attitude is capable of changing individual attitude, measuring group attitude is found to be useful.

iv) To modify the behaviours of the student into desirable ones, attitude scales are very useful.

v) Attitude scales also find an important place in understanding individual differences.

### 3:09:3:04 Limitations of Attitude Scales

i) There is also a view that the information obtained through attitude scales are not fully reliable. One may not express his true opinion, but say something else to project himself as a progressivist.



ii) Sometimes, a person may not be knowing his own attitude precisely.

iii) When faced with a real situation whether the individual will be firm in his earlier stated position is a big question mark.

iv) The five points given in the attitude scale are not at equal intervals, they have only imaginary boundaries.

v) Several statements which are not included in the attitude scale might be more appropriate.

vi) Attitude Scales indicate the trend in the outlook of the people (Progressive / Regressive) and not provide definite measures.

## 3:09:4 Motivation Scales

### 3:09:4:01 Meaning of Motivation

**Motivation is the process of arousing, maintaining and controlling interest in a goal directed pattern of behaviour**. Motivation is basic to all behaviours including learning. The success in life and learning depends on our motivation. It stimulates us and directs our behaviour.

Good motivation in any activity ensures that we develop an interest in the activity, feel an urge to do it, pay attention to it and the resulting performance is quick and efficient. On the other hand, if there is poor motivation, we feel the activity is forced on us against our desire. We may somehow do it or learn it in a haphazard way but our attention to the task will be minimum, mistakes will occur in plenty and performance will also be poor. The most important reason for the gap between pupil's potential and the current level of achievement lies in the area of motivation. According to **Crow** and **Crow, "Motivation is considered with the arousal of the interest in learning and to the extent is basic to learning".** An understanding of the nature of motivation, types of motivation and the innovative ability to make the best use of motivating influences to foster pupil learning are vital to teachers if they hope to get each pupil to make maximum use of his or her talents. Further it helps the teacher to know pupils' appetites and desires i.e. to become sensitive to pupils' needs.

Motivation in behaviour is shaped by three factors. They are : (i) Level of aspiration (ii) Level of expectation (iii) Mental satisfaction.

## 3:09:4:02 Measuring Motivation

Motivation is a psychological process, operating within the organism and as such it can not be directly observed. Hence it becomes necessary to measure one's motivation by using his / her reactions and self-reporting responses related to the cognitive, affective



and psychomotor domains. Generally to measure motivation a tool in the form of a questionnaire is used.

## 3:09:4:03 Some Examples for Motivation Scale

### 1. Miller's Motivation Scale

In Miller's motivation scale, there are 160 test items in the form of questions. These are arranged under three sub-headings (encouragement, self-satisfaction and social interest) and five divisions (innovativeness, creativity, effectiveness, cooperation and authoritative). Using Miller's motival scale one's level of general motivation can be assessed.

### 2. Measuring Tool for Students' Learning Motivation

Motivation can be broadly categorized as (i) intrinsic motivation and (ii) extrinsic motivation. There are separate tools for measuring these motivations. For example, the following motivation scale can be used to measure 'Students' learning motivation'. In this, a sample of 18 items expressing intrinsic and extrinsic motivation are given. This tool tries to test the five components of Student-learning motivation like interest, focus, self-confidence, perseverance and hard work, on a five point scale viz. 1) Very much agree 2) Agree 3) Not sure 4) Disagree 5) Very much disagree.

| S. No. | Test Items | Level on the Scale | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 |
| 1. | I prefer class work that is challenging so that I can learn new things. | | | | | |
| 2. | Compared with other students in this class, I expect to do well. | | | | | |
| 3. | I am so nervous during a test that I cannot remember facts I have learned: | | | | | |
| 4. | I am certain I can understand the ideas taught in the subjects. | | | | | |
| 5. | I have an uneasy, upset feeling when I take a test. | | | | | |
| 6. | I think I will receive a good grade in the class. | | | | | |
| 7. | Even when I do poorly on a test I try to learn from my mistakes. | | | | | |
| 8. | When I study for a test, I try to put together the information from class and from the book. | | | | | |
| 9. | I ask myself questions to make sure I know the material I have been studying. | | | | | |
| 10. | It is hard for me to decide what the main ideas are in what I read. | | | | | |
| 11. | When I study I put important ideas into my own words. | | | | | |
| 12. | I work on practice exercises and answer the questions given at the end of the chapter even when I don't have to. | | | | | |
| 13. | Even when study materials are dull and uninteresting, I keep working until I finish. | | | | | |
| 14. | I feel happy when I am able to understand all facts contained in the lesson. | | | | | |
| 15. | I put more efforts in learning, for the sake of getting good marks only. | | | | | |
| 16. | I often find that I have been reading for class but don't try to understand fully. | | | | | |
| 17. | When I read materials for this class, I say the words over and over to myself to help me remember. | | | | | |
| 18. | When I am studying a topic, I try to make everything fit together. | | | | | |



Among the motivation scales developed by psychologists, the following are worthy to mention.

1. **Smith's Inventory** containing 81 items for measuring one's motivation.

2. **Maslach's Inventory** containing 15 items that measures one's motivation.

3. **Weinstein's Inventory** with 77 items measuring one's motivation.

## 3:10 Measuring Interest

### 3:10:1 Meaning of 'Interest'

Getting involved oneself with a thing or activity shows his interest in that. There will be no compulsion in such involvement. Voluntary involvement is an indication of interest. Engaging one self in an activity in which he is interested is enjoyable to a person and he would like to spend more time in it. Interest and attention are mutually dependent.

Interest, in short is a behaviour orientation towards certain objects, activities or experiences. It is an expression of our likes and dislikes, or our attractions and aversions. The objects, activities or experiences which command our interests are stimulating, enjoyable and pleasurable whereas opposite is true in the case of our dislike.

### 3:10:2 Definition of 'Interest'

**Bingham** had defined interest as "a tendency to become absorbed in an experience and to continue it."

In a class, one student may be interested in writing short stories, another in making a model of an aeroplane and yet another in chatting with friends.

According to **Guilford** Interest is a generalized behaviour tendency of an individual to be attracted to a certain class of incentives or activities that are vocational in nature and to those whose broad meanings transcend vocations.

### 3:10:3 Interest Inventory

We are born not with some specific interests. We develop different types of interests due to the influence of environment. Generally tests to measure interest are developed in three broad areas. They are : (i) General Interests (ii) Vocational Interests and (iii) Educational Interests.

In interest inventory activities involved in different areas of interest are listed. From among the activities thus listed, the testee is required to choose and indicate those in which he / she has interest or 'no interest'. This enables the investigator to assess the individual's interest. This kind of interest measure-



ments are subjective in nature and are obtained by using interest inventories.

In the opinion of **Guilford**, there is close relationship between a person's interest and vocation. All the vocation could be included in seven types of interests listed below :

i) **Interest in Mechanics** : This includes activities that are mechanical or manual in nature with less emphasis on thinking.

ii) **Interest in Business** : This includes selling, business administration, marketing and development, social sciences and sensory satisfaction.

iii) **Interest in Science** : This includes activities like scientific investigation, scientific theory, mathematical concepts, laboratory work, logical processes, precision in detail, precision in carefulness.

iv) **Interest in Aesthetics** : This includes both aesthetic expression and aesthetic appreciation, enjoyment in drawing and painting, singing, dancing, undertaking literary work etc.

v) **Interest in Social Work** : This includes concern for the welfare of others, controlling others, office activities, responsibility, persuasion etc.

vi) **Interest in Clerical Work** : This includes such activities as office work, number manipulation, precision, exactness and physical activity.

vii) **Interest in Outdoor Activities** : They include outdoor activities like agriculture, construction, horticulture and forestry.

## 3:10:4 Some Important Interest Inventories

### 3:10:4:01 Kuder Preference Record

Kuder Preference Record, which could be administered to pupils between the ages of 9 and 16, was published in 1939 and revised in 1949.

This test is scored on the following ten scales : Outdoor, Mechanical, Computational, Scientific, Persuasive, Artistic, Literary, Musical, Social Service, and Clerical :

1) *Outdoor :* Indicates preference for undertaking work in agriculture, farming, forestry, building construction etc.

2) *Mechanical :* Indicates a preference for working with machines and tools.

3) *Computational :* Indicates a preference for working with numbers.

4) *Scientific :* Indicates a preference for discovering new facts and solving problems.



5) *Persuasive :* Indicates a preference for meeting and dealing with people and promoting projects for things to sell.

6) *Artistic :* Indicates a preference for doing creative work with one's hands. It is usually the work that has "eye appeal" involving attractive design, colour and materials.

7) *Literary :* Indicates a preference for reading and writing

8) *Musical :* Indicates a preference for going to concerts, playing instruments, singing or reading about music and musicians.

9) *Clerical :* Indicates a preference for office work that requires precision and accuracy.

10) *Social Service :* Indicates a preference for helping people

The blank contains 504 items, each offering three possible choices. The testee reads the three choices and indicates which he likes the most and which he likes the least. A typical item in it is:

a) Develop new varieties of flowers
b) Conduct advertising campaign for florists
c) Take telephone orders in florist shop

In this item, if an individual chooses 'a' as liked, he receives credit towards his score in the scientific and artistic areas. If he chooses 'b', is credited in the 'persuasive' area and if he selects 'c', he is credited towards the 'clerical score'. By adding the credits, a total score is obtained for each area of vocational interest. The scores obtained for the different areas are then converted into percentile ranks based on the responses of the so called "people in general".

### 3:10:4:02 Strong Vocational Interest Blank (SVIB)

**E.K. Strong** of Standford University, U.S.A. published his vocational interest blank in 1927 which was subsequently revised in 1951. In it there were listed 47 vocations for men and 28 vocations for women under 11 sections. The SVIB is available in separate forms for men and women from age 17 onwards. Each inventory contains 400 test items dealing with likes and dislikes, various objects, occupations, school subjects, amusements, activities, and social contacts in daily life (like meeting senior citizens, army officers, social workers, workers involved in dangerous tasks etc.) are given in pairs. From each pairs the testee has to choose one that reflects best his present characteristics and abilities.



The main purpose of Strong's Vocational Interest Blank is to compare how far the interests and preferences of the testee agree with those of successful persons in specified occupations.

### 3:10:4:03 Strong Campbell Interest Inventory (SCII)

**David P. Campbell** modified Strong Vocational Interest Blank and published it in 1974 with 325 items under seven sections. In the first five sections the testee records his preferences by marking any one of the responses viz. 'Like', 'Indifferent' or 'dislike'. The items in these five sections fall into the following categories (i) Occupations (ii) School Subjects (iii) Activities in day to day life (e.g. making a speech, repairing a clock, raising money for charity) (iv) amusements and (v) day-to-day contact with various types of people (e.g. senior citizens, army officers, people involved in dangerous tasks etc.). The remaining two parts require the testee to express a preference among the triads (three alternatives given in an item) given in the form of descriptive statements by marking 'Yes' / 'No' / 'Not sure' for each item. One of these two sections, has items comparing things and men related interests and in the other the comparison will be about characteristics and abilities. Marks from each section will be ranked to find out his field of interest. Though there is no time limit for this test, it could be finished in 30 to 60 minutes by most testees.

## 3:10:4:04 Thurstone's Occupational Interest Schedule

This was developed by **L.L. Thurstone.** In this schedule the testee is asked to express his preferences for different occupations. The occupations are given in pairs. The paring is done in such a way that a particular occupation placed first in a pair gets placed second in another combination of two occupations. Thus a large number of pairs are developed for comparing the different occupations. The testee is asked to check all the pairs of occupations given, to indicate his preferences. In each comparison, the testee is to assume that there is no difference in income or prestige. While comparing, for each pair of occupations, the testee is to record his preference as follows :

i) Circle the occupation which he prefers, out of the two.

ii) Circle both, if both the occupations in the pair are equally liked.

iii) Mark XX (i.e. crossing out), if both the occupations in the pair are disliked.



*Example :*

| | PSI | BSI | CI | BI | EI | ... | ... |
|---|---|---|---|---|---|---|---|
| PSI | 1. Physicist<br>2. Engineer | 1. Physician<br>2. Physicist | 1. Auditor<br>2. Chemist | 1. Banker<br>2. Machine Designer | 1. Campaign Manager<br>2. Civil Engineer | ...<br>... | ...<br>... |
| BSI | 1. Civil Engineer<br>2. Anatomist | 1. Physiologist<br>2. Zoologist | 1. Statistician<br>2. Botanist | 1. Business Manager<br>2. Physiologist | 1. City Mayer<br>2. Biologist | ...<br>... | ...<br>... |
| CI | ... | ... | ... | ... | ... | ... | ... |
| ... | ... | ... | ... | ... | ... | ... | ... |
| ... | ... | ... | ... | ... | ... | ... | ... |

[Note :PSI refers to 'Physical Science Interest'

BSI refers to 'Biological Science Interest'

CI refers to 'Computational Interest'

BI refers to 'Business Interests'

EI refers to 'Executive Interest']

In the revised interest schedule there are 100 squares (ten rows and ten columns), each containing 2 occupation names of which the examinee may indicate his preference, or may accept or reject both. Ten interest groups are represented : (i) Physical Science (ii) Biological Science (iii) Computational (iv) Business (v) Executive (vi) Persuasive (vii) Linguistic (viii) Humanitarian (ix) Artistic and (x) Musical. For every preference indicated, one mark is given.

Possible total score for each of the 10 interest groups is from 0 to 20. The obtained scores are plotted as a profile. This revised interest schedule is widely used in vocational guidance.

### 3:10:4:05 Lee-Thorpe Occupation Interest Inventory

The Occupation Interest Inventory developed by **Lee** and **Thorpe**, consisting of 240 items, attempt to measure one's occupation related interest. In this test, the respondent is required to select and tick (✓) one of the statements in each pair of two descriptive statements. Test items, arranged under six areas of occupation, describe the different kinds of work related to the concerned area of the occupation. The areas of occupation contained in this occupation interest inventory are :

a. Social and Personal

b. Related to 'Nature'

c. Mechanical

d. Economy

e. Arts

f. Science

## 3:11 Types of Test Items

Generally a written test may consists of different kinds of test-items. Test-items differ on the basis of



the answers, they require. Test-items could be contained in two broad types. They are :

#### i) Free or Uncontrolled Response Type

For this kind of test-items, testees can provide answers in their own way. This type includes 'Essay Type' items (requiring 3 to 4 pages of answer) and 'Short Answer Type' items (requiring answers in about a paragraph or five lines).

#### ii) Fixed / Controlled Response Type

In this kind of test-items, testees can not provide answers as they like; their answers are controlled by the test-items. Objective type questions and very short answer type questions (requiring a word, phrase or one / two sentences as answers) belong to this category.

Let us next discuss about different kinds of 'Objective type' test-items in the pages to follow.

### 3:11:1 Objective Type Test-items

Objective type test-items could be classified into the following two kinds: (i) Recall type and (ii) Recognition type.

Multiple choice test-items, Completion type items, Matching type items, True/False items, Find the odd one from among those provided etc. are included in the Objective type test-items. Among these completion type items and very short answer / one word answer types belong to the recall type; the rest are selection type test-items.

Let us discuss about each of them in detail.

### 3:11:1:01 Multiple Choice Type Test-Items

In this type of Test-items, the testee has to select one among the answers provided, which he considers the most appropriate and offer it as his response. For every test-items, three to five alternatives (answers) may be given. In this type of test-items, the stem may be an incomplete sentence, an interrogative sentence or containing one or two sentences, according to the requirement.

**Illustration :**

Which among the following can not pass through vacuum?

a) Light b) Sound c) Magnetic field d) Electric field

In this, the interrogative sentence, 'which among the following can not pass through vacuum' is the **stem;** those given against a), b) c) d) are known as **choices or alternatives.** Here (b) is the correct answer; this is called the **'Key'** and all others are called **'Distractors'.**



Though preparing multiple choice test-items are difficult, they could be used to test all levels of instructional objectives (i.e. knowledge, understanding, application, analysis, synthesis etc.). Therefore among the objective type test-items, multiple choice items are used more often. Further, the chances are very low for answering multiple choice test-items blindly, without any thinking.

### 3:11:1:02 True/False Type Items

In this type of test-items, the testee has to read the given statement and indicate what it conveys is true or false by putting 'T' or 'F' accordingly in the bracket provided for. The big limitation of this type of item is, that the chances are high for students, even without knowing the subject content, to guess and get full marks for his response; as there are two alternatives only, the chance of one getting his / her guess right is 50%. Further this kind of test-items can test only 'Knowledge' level instructional objectives.

### 3:11:1:03 Matching Type Items

In matching type items, there will be two columns - (i) Stimuli / Premises and (ii) Responses. This type of items is only an alternative form of multiple choice questions. Using this type of test-items, definiteness of students' factual knowledge could be tested. In this type of test-items, facts related to each other will be provided randomly in two columns (stimulus column, response column) and students are required to match every given stimulus with the most appropriate among the responses given. This type of test-items bring out students' ability to **recognize** the correct pairs.

**Illustration**

Match those given in column 'A' with those in column 'B' of the following test-item.

| S.No. | Column 'A' | Column 'B' | |
|---|---|---|---|
| 1. | Thorndike | (a) Operant conditioning | ( ) |
| 2. | Pavlov | (b) Insight learning | ( ) |
| 3. | Kohler | (c) Trial and Error learning | ( ) |
| 4. | Bruner | (d) Concept learning | ( ) |
| | | (e) Classical conditioning | ( ) |

### 3:11:1:04 Very Short Answer / One Word Answer Type

This type of questions have answers of very small size. These are simple recall type test-items. Answer for this type of questions may be a word, phrase or one or two sentences. This kind of test-items belong to 'Supplying Type'.



**Illustration**

i) Which is the capital of India?

ii) What is the name of a quadrilateral having all sides equal?

### 3:11:1:05 Finding the Odd one from Among those Provided

In this kind of test-items, except one of the given words, all others are related to an attribute or a concept. The testee has to identify the one which stands unrelated.

**Illustration**

From among those given, find the odd one a) Rectangle, b) Circle, (c) rhombus, d) Parallelogram.

The correct answer for the above item is 'b) circle as the other three are four sided figure. For a circle there is no definite side. This type of test-items belong to the 'Recognition' type.

### 3:11:2 Merits of Objective Type Test Items

1. As there is only one correct answer for each of the objective type questions, there is no room for individual's score being affected by the scorer's mood and many examiners valuing the answer scripts of students etc.

2. Using the scoring key, answers could be scored mechanically very fast and easily.

3. Scoring is free from personal-bias of examiners.

4. It is very easy for pupils to understand and answer this type of test-items.

5. This type of test-items facilitate better content coverage across the whole syllabus as more number of test-items could find place in the question paper i.e. comprehensive evaluation of students' achievement becomes possible.

6. The chances of students expecting some important questions or relying on luck could be avoided by making use of this type of questions.

7. This type of test-items do not require more time to answer.

8. It becomes possible to conduct the examination for thousands of students on a single day and announce the results shortly.

9. Each objective type question is based on a clearly stated specific learning objective.

10. This type of test-items discourage cramming and encourage thinking.

11. Objective type tests are more reliable than other forms.



### 3:11:3 Limitations of Objective Type Questions

1. By using objective type test-items, we can not test the ability of students to arrange their ideas coherently.

2. It is not possible to evaluate how well students express their ideas.

3. Chances are more, for guessing the probable answer for the test item; i.e. this type of test-items evaluate superficial knowledge of students.

4. It is very difficult to prepare objective type test-items.

5. Only well-experienced teachers can prepare good objective type test-items.

6. More importance is given only for subject content i.e. this type of test-items do not encourage the skills like elaboration of a concept, imagination, presenting ideas in an innovative way etc.

7. Objective type tests neither promote thinking ability nor language ability.

### 3:11:4 Short Answer Type Questions

Questions which require answers in not more than three or four lines / 40 words are called 'Short Answer Type'. Here students are required to think and express the correct answer precisely. Using this type of questions, it is possible to prepare a question paper covering the entire content areas to be tested. Further evaluation of the answers of short answer questions is more objective than essay type questions and hence has more reliability. However the objectivity and reliability of short answer questions are comparatively less than objective type test items.

### 3:11:4:01 Merits of Short Answer Type Questions

i) Students are required to express their answers precisely and specifically, without beating around the bush.

ii) Scoring of answers is little more objective and reliable.

iii) It is possible to prepare the question paper covering the entire content areas to be tested.

## 3:11:5 Paragraph Type Questions

Paragraph type questions require answers not exceeding 10 lines / 100 words. The answers of this type of questions are to some extent subjective in nature, This type of questions are highly useful in testing specific learning outcomes. Preparing questions of this type is easier than that of objective type test items.



## 3:11:6 Essay Type Questions

For this type of questions, students have to give a detailed written response, running into 4 or 5 pages (a maximum of 600 words), arranging the answers in paragraphs and with sub-titles. For answering essay type questions, students have to recall, what they have learned and retained in memory.

Essay Type questions test students' ability to think and present their answers elaborately and logically. For this they need to gather all pertinent information related to particular issues / problems and present the appropriate answers by arranging the collected information coherently. Other type of test-items do not provide such opportunities to students.

### 3:11:6:01 Merits of Essay Type Questions

i) Essay type questions promote in students the ability of recalling selective information from among those they have learned.

ii) This type of questions provide opportunities for students to express freely their own ideas, thinking and the elaborate knowledge they have.

iii) They promote in students the ability to elaborate ideas as well as compare and contrast them.

iv) They provide opportunities for students to establish relationship between facts, interpret concepts and principles and solve problems by using the appropriate method and means.

v) They provide students the opportunities for creative thinking and making use of new approaches while presenting their answers. Further they promote language ability and presentation skill in students.

vi) They develop good study habits in students and cultivate the skill of note-taking from good reference books.

vii) They develop in students abilities like critical reasoning and systematic presentation of their ideas in a well organised manner.

viii) Chances of guessing the answers as well as copying from others in examinations will be minimum.

ix) They avoid students memorizing the answers and reproduce them verbatim as well as answering by mere guessing.

x) It is very easy to prepare an essay type question.

### 3:11:6:02 Demerits of Essay Type Questions

1. The question paper containing only essay type questions have very low content validity since only few essay type questions can be asked in an examination which can not cover the entire content areas to be tested.



2. Essay type questions require more time for answering.

3. When the answer for an essay type question provided by a student is valued by many examiners, too much of variation is found among the marks awarded by them. Apart from this, when the same examiner is asked to value the same answer script at different points of time, different marks are given by him. These point out that essay type questions have poor reliability.

4. Besides low validity and reliability, essay type questions lack objectivity too. Students with good language ability can score good marks inspite of not being good in their knowledge of the subject content.

5. The quality of the hand writing as well as length of the answer rather than the depth influence the examiner.

6. Examiner's personal bias, and his mood too affect the marks awarded for the answer to an essay type question.

7. Essay type questions give importance for rote learning and the ability to retain and recall the answers.

## 3:12 Principles for Constructing Test-items

Generally the following criteria are to be followed in developing any type of test-items.

a) Each test-item should aim to test one or more instructional objectives i.e. test-items should have content validity.

b) Test-items should be constructed on what the testees should know and are able to make meaning of it.

In addition to the above, there are some specific principles for developing each type of test-item, the details of which are presented below.

### 3:12:1 Principles Related to the Construction of Multiple Choice Test-items

#### I Principles related to the 'Stem'

i) Setting the problem related to the objective, in the stem of the test item.

ii) Stating the stem in clear, definite and explicit words.

iii) Avoiding excessive and non-functioning words.

iv) Avoiding double negative words and phrases.

v) Stem should be stated in grammatically correct language.



vi) Avoiding in the stem any clue to choose the answer.

vii) Avoiding negative statements in the stem as far as possible.

#### II Principles Related to the 'Choices'

i) Presenting atleast four choices

ii) Using attractive and plausible alternatives as distractors.

iii) Avoiding responses like 'all the above' and 'none of the above'.

iv) Making correct responses appear at random order.

v) Alternatives provided should fit in well with the stem grammatically.

vi) Alternatives provided should be mutually exclusive.

vii) Arranging alternatives one after the other, in numerical order.

### 3:12:2 Principles Related to the Construction of a 'Matching' Test-item

1. Items in the matching question should be arranged in two columns, one for the 'premises' and another for the responses.

2. Generally the premises should find place in the left side column and responses on the right side column.

3. Number of responses given should be one or two more than the number of premises given.

4. The premise and response of the matching pairs should not find themselves just opposite to one another.

5. Items chosen for 'premises' and 'responses' of a matching type question should belong to the same topic or content area i.e. homogeneous materials should be used to form the items of a matching question.

   (e-g): i) Theories - Those who propounded

   ii) Defense Mechanisms - Acts / behaviour

   iii) Nomenclature of bodily characteristics - Personality types

6. All the items for a matching type question should find place on the same page.

7. Number of 'premises' in a matching type question should be counted as the number of test–items.

8. Proper instructions should find place at the top of the matching type question.

## 3:12:3 Principles Related to the Construction of 'Completion Type' Test-items

1. In this type of test-items, care should be taken to see only one answer (a word or phrase) fit in well in the blank space provided.



2. The blank provided in the question should imply a specific idea that is to be assessed.

3. A blank should not find place at the beginning of the statement given.

4. Statements should not be taken directly from the text book and use them as such for forming the completion type test-items.

5. Clue for the answer, should not be contained in the statement, either through the grammatical structure or informational structure.

6. When the answer is to be expressed in numerical units, then the clue for the precision of the number should be indicated.

   (e-g): Instead of forming a test-item as

   "Thirukkural contains ___ 'kurals' totally",

   it is better state it in the following manner :

   "Number of 'Kurals' contained in Thirukkural is _____

7. Too many blanks are not desirable in an item.

## 3:12:4 Principles Related to True / False Test-items

1. Given statements should be either 'fully true' of 'fully false' if they are to be judged as true or false.

2. Given statement should contain only one idea except in the case of cause-effect relationship.
3. Avoiding the usage of words like 'May be', 'Some' 'many' etc.
4. Avoiding long complex sentences.
5. Avoiding the use of negative phrase (like ill-logical, disrespect etc.) and double negatives in the statement given.
6. Statements should be formed based on well established facts.
7. If opinion is used, its source should be attributed (E-g) According to Skinner, no stimulus is capable of eliciting a unique response from an organism.
8. True and False statements given should be approximately equal in number and randomly arranged.
9. The correct responses should not follow a pattern.
10. Use of words like 'All', 'Always', 'none of them', 'only' etc. should be avoided in the statements given, since they sometimes may provide the clue for the correct answer.



### 3:12:5 Principles Related to Constructing Short-answer Type Questions

i) Questions should be framed with clear and meaningful words.

ii) Questions should be so structured that the acceptable answers are in few sentences or listing of few items.

iii) When questions involve computational problems, it should be clearly stated that answers are to be expressed in proper units and the degree of precision required for the answers.

### 3:12:6 Principles Related to Constructing Essay Type Questions

i) Words and language style used in stating the questions should be simple and easy to understand for students.

ii) Questions should test the ability of students to analyse, synthesize and evaluate.

iii) The size of the answers expected for the essay questions should be clearly mentioned (like not exceeding 4 pages or 600 words).

iv) Weightage to be assigned (in terms of marks) for essay type question should be determined in advance and stated clearly in the question paper.

v) Essay type questions are to be used only for the purpose it serves best.

## 3:13 Major Issues in the Assessment of Learning

Major issues in the assessment of learning are :

i) Commercialization of assessment

ii) Poor test quality

iii) Domain dependency

iv) Measurement issues

v) System issues

### 3:13:1 Commercialization of Assessment

Commercialization of education has been fairly a recent trend in India. Its main emphasis is to make education business oriented, earning maximum profit for the investment put in. Privatization of education has resulted in the mushroom growth of self-financing private educational institutions and they are collecting hefty tuition and other fees, in the name of providing quality education. Schools instead of promoting all round development in pupils, fostering true knowledge, critical thinking, creativity and social awareness etc. they have nowadays become commercial establishments making money in everything like learning materials such as text books, note books etc. uniform



dress materials, shoes, transport, food items etc. and forcing students to avail of these things tagged with school label. They charge additionally for value added programmes provided, such as 'Yoga', training in spoken English, martial arts (like Karate), job-oriented courses like Computer programming, travel and tourism, fashion designing, learning an additional language etc.

In the assessment of learning achievement of student, instead of giving importance for **'process oriented assessment'** which mainly focuses on formative evaluation of learning, more emphasis is given for **'product-oriented assessment'**. As a result of this getting high marks in the school final examination has become the goal of school education today. Instead of giving emphasis for students learning with comprehension, schools have become mere **'Knowledge shops'** that serve to provide students, training in rote learning by memorizing the subject content and efficiently retrieving them verbatim.

Teachers provide projects and assignments to students, on various topics in the subject content to promote in them problem-solving skills, the ability to gather relevant information for completing a given assignment, the ability to think and experiment with innovative ideas and the ability to prepare simple learning aids and materials that make them understand the laws and principles they learn. However students, instead of involving themselves in these activities and enhance their learning, they seek the help of their parents and get their projects and assignments completed. Many commercial shops have sprung up, to produce things that are required by students and make them readily available for sale. Among the engineering college students, the recent trend is to place order and purchase their project as finished product and submit the same for assessment by examiners. In teacher education colleges too, B.Ed.. students purchase improvised apparatus, charts and diagrams from **'readymade shops'** and submit them in their practical examination. These 'readymade shops' have now gone up one step further to make available these teaching-learning materials on rent for a few days.

Some publishers have indulged in selling printed practical records for laboratory work, micro-teaching record, camp record, psychology practical record, SUPW record etc. with **'recorded readings'**. B.Ed. students are tempted to purchase these records and submit them for assessment in the practical examination.

Putting in brief, at present students instead of believing in working hard to learn with understanding,



by gathering relevant information from various sources through visiting libraries, surfing websites in the internet etc., focus their attention in getting their work done on behalf of them by spending money and aim to score high marks in examinations.

The main reason for learning activities getting commercialized is, according undue importance for summative evaluation of learning achievement rather than giving importance for formative evaluation of learning i.e. without evaluating the 'process of learning, focussing much on the product of learning is the real cause for degradation in the assessment of students' learning achievement.

### 3:13:2 Poor Test Quality

Tests serve as important tools in the assessment of students' learning achievement. Only if tests employed are of high quality, they could help the teacher to find the real strength and weaknesses of students, based upon which he could provide them proper feedback to improve their learning. Following are the important characteristics of a good assessing tool like test

i) Validity

ii) Reliability and

iii) Objectivity

If all the test items in a test, measure what the test intends to assess, then the test is said to have high amount of validity. If the test items do not sample out adequately from among the content elements that assess the attributes to be tested, then the quality of the test is poor. Putting it in brief, the **'Content Validity'** of a test could be increased, by providing due importance to all the content areas that are included for the test and developing suitable number of test items across the entire range of content areas meant for the test.

Test items, besides being stated with clear and unambiguous words, should also indicate how much is the expected size of the student - response. For example, instead of framing a question as 'What are the reasons for the poor quality of a test?' it is better to design it as 'Mention any three important reasons for the poor quality of a test', so that there will be no chances for variations in scoring the student responses i.e. objectivity in scoring will be high. As the objectivity of a test increases, along with it, its reliability too increases.

The assessment of answer scripts of a test whenever be taken, if there is not much variations in the marks awarded, then the test is considered to be highly reliable. If there is no variation in the marks



awarded, whoever scores the answer scripts of a test, then its objectivity is very high.

Most of the achievement tests conducted in our schools at present are of poor quality. Following are the reasons for this :

i) Test items do not evaluate the instructional objectives which they presume to do.

ii) Test items allow chances for students to guess the answer.

iii) Test items having more than one correct answer.

iv) In some of the multiple choice test items, none of the alternatives given, suitably fits in for the correct answer.

v) Questions are being vaguely stated i.e. students can not understand what should be the size of the response expected and for aspects of the answer more emphasis should be given. Assessment of the answer scripts of tests containing such vaguely stated questions will be highly subjective; the reliability of such tests too will be low.

Achievement tests, besides assessing students' learning attainments, also attempt to provide feedback enabling students to improve their learning. Hence, achievement tests with poor quality not only fail to assess students' true learning achievement, they also do not contribute to improving student-learning. Most of the achievement tests, often contain test-items based on the instructional objectives 'recall' and 'recognize' the content elements which help to assess students' ability to memorise and retrieve; they do not assess the higher order mental skills. Inservice training should be given to teachers with regard to 'Developing different types of test-items' and 'Construction of a good achievement test' so as to improve the quality of achievement tests used in-our schools.

### 3:13:3 Domain Dependency

Whatever may be the learning, it is based on three domains - (i) Cognitive domain, (ii) Affective domain and (iii) Conative or Psychomotor domain.

Cognitive domain includes mental activities that help to construct one's knowledge. Emotional development which is related to appreciation, attitude, values etc. of an individual denotes the affective domain learning.

Skills based on neuro-muscular acts of our bodily organs are included in the psychomotor domain.

Among these three domains, cognitive domain is considered to be the major one, in the instructional process. The maturity achieved in the learning related to the cognitive domain makes it possible to achieve



the expected learning outcomes in the other two domains. viz. affective and psychomotor domains. Hence cognitive domain is given more importance than the other two in the teaching-learning process.

According to **Benjamin Bloom,** Cognitive domain consists of six levels - (i) knowing, (ii) understanding, (iii) applying, (iv) analysing, (v) synthesizing and (vi) evaluating. These abilities starting from 'knowing' are hierarchically arranged in the ascending order, on the basis of their development and difficulty level. These form the six aspects of knowledge development and guides the assessment of the knowledge attainment of learners at the end of an instructional period. These six aspects helping to assess learning outcomes, should be kept in mind, while planning to construct an achievement test.

'Knowing' is related to retention in memory of what are learned. Retrieving those recorded in memory is of two types - (i) 'Recalling' and '(ii) 'Recognising'.

The cognitive skill of 'understanding' involves, making use of the knowledge and skills one has acquired in a new situation. Behavioural elements like 'Giving reason for', 'Establishing relationship', 'Infer', 'Problem-solving' etc. are included in the cognitive skill of 'Application'. The cognitive skills 'Analysis', 'Synthesis' and 'Evaluation' are successive higher order mental skills, following the skill of 'Application'.

The major problem involved in the assessment of learning achievement is that most of the tests contain test-items that test 'knowledge' and 'understanding' of students; they do not attempt to test higher order mental skills like 'application'. Hence, the validity of such tests is usually low; these tests lack the ability to assess the true learning proficiency of students. Achievement tests should be so constructed that they contain atleast 20% of the total test-items that test higher order mental skills.

Instructional objectives in the affective domain, beginning with developing 'appreciation' gradually reaches the highest level of 'character formation', through the intermediate levels of developing 'interest', 'attitude' and 'value' in the learners. Learning achievement in the affective domain is to be measured by using aptitude tests, attitude scales, interest inventories etc. Achievement tests are not appropriate for assessing the affective domain instructional objectives.

Skills learned by students are to be assessed using the **behavioural elements** like 'Perception', 'Imitation', 'Manipulation', 'Precision', 'Articulation' and 'Naturalization'. Practical tests which evaluate



students' skills of drawing, marking the particular points or parts in a given diagram, organizing a laboratory experiment efficiently, recording the measurements accurately and expressing them in proper units etc. should also be given importance in the assessment of learning proficiency achieved by students.

As the three domains of learning are different, for assessing students' learning proficiency, employing different methods of assessment becomes indispensible. In assessing student-learning, in addition to written tests employing oral tests, observation of performance, assessing the ability to work with others as a team etc. too become necessary. Briefly stated, formative and summative tests should form integral parts of Continuous and Comprehensive evaluation of student learning.

## 3:13:4 Measurement Issues

**Measurement in education** means finding out students' learning achievement in the cognitive, affective and psycho-motor domains. It involves four activities - (i) designing opportunities to gather evidences, (ii) collecting evidences, (iii) interpreting the collected evidences and (iv) acting on the interpretations. In the assessment of learning there are two aspects - (i) Formative evaluation and (ii) Summative evaluation.

### 3:13:4:01 Issues in Measuring the Affective and Psycho-motor Domain Objectives

Besides the written tests used to measure the cognitive domain objectives like 'knowledge', 'understanding' and 'application', techniques to measure the affective domain objectives like attitude, involvement / interest etc. and different skills of the Psycho-motor domain are also necessary in the formative assessment of students' learning. In assessing the affective and psycho-motor domain objectives, teacher's observation finds an important place. When there is no proper teacher's observation of student learning, it leads to wrong interpretation of the measurements obtained.

**Problems in Using observation as an assessment technique**

i) The personal likes and dislikes of the observer and his own limitations will affect the quality of observation.

ii) Only expressed behaviour could be observed; the inner feelings can not be found out.

iii) Observer could obtain information related to one aspect of student behaviour only.



iv) Recording may not be done on the spot; the data recorded may not be accurate.

v) The presence of the observing teacher may be a reason for students' natural behaviour not getting expressed.

vi) Observation requires more time, more patience and keen insight.

Creating suitable opportunities for observing students' behaviour and giving proper interpretation for the data obtained in the observations are the major issues in the formative assessment of student-learning in the affective and psycho-motor domains. Teachers' tact and experience may help to a great extent to manage these issues.

### 3:13:4:02 Issues in Measuring the Achievement of Cognitive Domain Objectives

Written tests find an important place in measuring the learning achievement in the cognitive domain objectives. Test items that are included in written tests should be arranged sequentially and related to the different aspects of learning. Limitations found more often in test-items of written tests are :

i) Test items are not found spreading across the content areas of the test; but amassed in one or two content areas only.

ii) Questions are not formed with simple and unambiguous words.

iii) Including questions that test 'knowledge' objective more in number; not giving much importance to questions that test higher order mental skills like 'application'.

iv) Not providing appropriate directions regarding the size and nature of the answers expected for 'Essay' and 'Short Answer' type questions.

v) Without preparing the rubrics, start evaluating the answer scripts.

vi) Questions having short answers are framed as 'essay type' questions.

vii) While evaluating answer scripts, scorers giving room for 'generosity error', 'halo effect', 'average error' etc.

### 3:13:4:03 Awarding Marks in the Measurement of Students' Learning Achievement

Students' learning achievement cannot be measured directly and accurately as we could measure their external attributes like height, weight etc. Hence it may not be appropriate to use a measuring scale for evaluating students' learning achievement. Learning achievement of a student is to be understood



by comparing it with that of the fellow students (or peers). Therefore it will be more appropriate to **grade** the learning achievement of students on a five point or seven point scale such as 'Very Good', 'Good', 'Fair/ Average', 'Below Average' and 'Poor'. For such evaluation of students' answers, development of **'rubrics'** and not **'a scoring key'** is required.

### 3:13:4:04 Use of Standardized Tests in Measuring Proficiency Achieved in Learning

In using standardized tests to measure learning achievement of students, we should see whether 'standardization' is done properly as otherwise, truthfulness and reliability of the obtained measurements may get affected.

### 3:13:4:05 Issues Related to other Factors in Measuring Learning Achievement

i) While measuring the learning achievement, if data related to all the aspects of learning are not available, then the obtained measurements and the interpretations based on them get affected.

ii) Using the data obtained by assessing students' learning achievement, without giving proper weightage for different aspects of learning, then our assessment of students' learning achievement may become erroneous.

iii) While assessing student learning, some extraneous factors may produce positive or negative effects on our measurements.

iv) In some unavoidable situations, where we have to use certain learning achievement measurements, they may produce significant influence in our assessment.

## 3:13:5 Problems Related to Evaluation System

If we consider 'Education Process' as a system, then 'Learning Assessment' is a sub-system of it. A system can function efficiently, only if all its sub-systems function well. According to **Gitomer** and **Dusche** (2007) "In the 'Learning assessment System', there are two sub-systems called 'Formative assessment' and 'Summative assessment'. Formative assessment consists of oral tests, short answer assignments, assessment of student - participation in the classroom learning activities etc. In the summative assessment, written tests find an important place. If these two types of assessments do not function independently as well as helping each other, then the system 'Learning assessment' will get plagued with problems".

The main issues that may arise in an evaluation system are :



i) Evaluation system that completely relies on written examinations and the problems related to it.

ii) Problems related to the use of rubrics.

iii) Problems related to the evaluation of student performance

### 3:13:5:01 Written Examination based Evaluation System

A learning assessment system that relies only on paper-pencil tests can not implement formative assessment effectively. Formative assessment will largely remain weak, if importance is not given for assessment techniques like oral tests and teacher's observation which could evaluate students' participation in day-to-day classroom teaching-learning process.

Written examinations conducted at the end of the instructional period or each semester / term, may provide for certification of proficiency attained in learning but of less help in improving student-learning. Further, written examination question papers which lack in validity and reliability, having test-items not well spread over all the content areas and containing more number of items that test mainly 'Knowledge objective', will not help much for assessing student learning truly and accurately.

### 3:13:5:02 Problems Related to Preparing Rubrics

Preparing rubrics is easy for assessment of student learning through observation like assessment of physical skills developed in students, assignments completed by them etc. But preparing rubrics is difficult for written tests, particularly for 'Essay type questions'. If rubrics are not developed by the teacher by having a detailed discussion with students, they will lack transparency leading to problems in assessing students' learning achievement.

### 3:13:5:03 Problems Related to Students' Performance

Unless rubrics are prepared and used in the assessment of physical skills and experimental skills involved in carrying out laboratory experiments, like fixing and handling apparatus properly, observe and record the readings accurately etc. problems will crop up with regard to the assessment of students' performance.

Problems related to the formative and summative assessments are with regard to the **internal components** of the evaluation system. Problems related to the external components of the evaluation system are those with regard to the method of



conducting the assessment tests (i.e. those related with conducting it within the prescribed time-limit, students copying answers from others, impersonation of candidates in the examination hall etc.)

## 3:14 Reforms in Assessment

The present system of examination encourages students to memorise things that are to be learned, retain them in memory and retrieve them efficiently rather than testing their meaningful understanding of the subject content. Classroom teaching-learning process itself has reduced to teachers providing information related to the subject-content and students absorbing and memorizing them. Teachers focus on covering the content areas through classroom instruction and students are more concerned with passing the examinations. Question papers mostly lack validity, reliability and objectivity. Non-cognitive learning outcomes are neither tested continuously nor comprehensively. But educational researches advocate the assessment of student performance and comprehensive evaluation of learning achievement by using many types of assessment methods. **Bringing in appropriate changes in the evaluation system facilitating comprehensive assessment of students' learning achievement by adopting multi-dimensional assessment of learning is known as 'Reforms in assessment'.**

From Mudaliar Education Commission (1952-54) to National Policy on Education (1986), various Committees and Commissions have suggested various examination reforms, of which the following are the important ones.

i) Annual examination system is to be changed into semesterwise examination system. In the annual examination system, students pay less attention to their studies and start working hard at the year end to get success in passing the examination. On the contrary, in the semester pattern of examination, students are kept busy in learning throughout the academic year.

ii) Apart from conducting examinations at the end of each semester, continuous internal evaluation tests are to be held and suitable feedback provided to students then and there, to improve their learning.

iii) Through 'Continuous and Comprehensive Evaluation' (CCE) System, both scholastic and non-scholastic learning achievement of students are to be assessed.

iv) Conducting quizzes, providing assignments and projects periodically, asking for the submission of 'student learning portfolio' from each student could be undertaken in the continuous internal evaluation programme.



v) Along with written examinations, oral tests and practical examinations too, to be conducted.

vi) Students' achievements are to be continuously monitored by maintaining proper **'Cumulative records'.**

vii) In written examinations subjectivity could be minimized by reducing the number of essay type questions and increasing the number of objective type test items.

viii) Test items, instead of giving importance for testing the ability of students to express from their memory what they have acquired by rote learning, should often test higher order mental skills like understanding, application, analysis, synthesis etc.

ix) Instead of marks, grades should be adopted to represent students' learning achievement.

x) Before teachers start evaluating students' answers, assignments and performance, 'rubrics' should be prepared and made available to students. By this students will be able to know how they are going to be evaluated and graded for their achievement i.e. the evaluation should be more transparent.

xi) In accordance with the assessment system, instructional methods and materials too should be modified.

## 3:15 Open Book System of Assessment

Open book system of assessment allows students to take texts or resource materials, reference books, notes prepared by them etc. into the examination hall, use them to identify the relevant answers for the questions asked in the written examination, reflect over them and present their final answers. This kind of assessment system tries to test the ability of students to find and apply the information and knowledge, statistical data, precedents, citings, quotes etc. from various sources, in preparing their answers, within the given time limit.

### 3:15:1 Two Forms of Open Book Examinations

Open book examinations could be implemented in the following two forms.

**i) Traditional Method**

In this method, similar to conducting usual examinations in schools, students are required to write answers for the question paper given to them, in the presence of the invigilators in the examination hall and submit their answer scripts within the time-limit for the examination; but the only difference is, students are



allowed to refer texts and other resource materials which they have taken to the examination hall before the start of the examination and prepare answers. The essence of this method is students are required to take their seats in the examination hall and finish answering the question paper by referring texts and other resource materials, within the given time limit.

**ii) Answering the test from home**

In this method students could take the question paper to home, prepare their answers and return the answer script, the next day.

### 3:15:2 Kinds of Materials Used in Open Book Examinations

Text book, reference books, notes prepared by respective students etc. are usually allowed for students to refer and prepare their answers in the open book examinations; but guides available in the market are not permitted. Generally, the number of reference materials used in the open book examinations are not restricted; the only restriction imposed is that students should prepare answers individually of their own, without seeking the help of others.

### 3:15:3 Kinds of Questions Finding Place in Open Book Examinations

Test-items in the open book examinations do not test students' memory i.e. ability to remember and recall. They test the ability to find and use information for problem-solving and to present well-organized arguments and solutions.

In the open book examinations no direct questions are asked either from a topic or unit. As they demand students to compare, contrast, differentiate, explain the contradictions among the laws, the ability to refer some books or resource materials, spot the relevant information and apply them appropriately is needed very much. Since the time limit for answering the question paper is fixed, unless students have already read the text books or reference materials once or twice, it is very difficult to locate the relevant information from among the books at hand.

Generally questions in open book examinations may be the essay type questions or involve problem-solving or presenting arguments for and against to a solution suggested for a problem. The style of questions, depends on the teacher who frame them or the educational institution that sets the examination.

## 3:15:4 Misconception About Open Book Examinations

**1. Very easy to succeed**

Since no direct questions are asked, the information present in the books cannot be presented



as such while answering the questions. Questions test the ability of students to identify the right book among those permitted to view that contains the required information, locate in it the relevant information and apply it intelligently in presenting the answer.

**2. No previous reading of books is required**

It is a misconception that without any previous reading, students can directly take on the written examinations. Only if books are read already, it is possible to locate the relevant required information quickly and based on them prepare the answers within the given time-limit.

**3. Using the Information from books as such is possible**

Information located in books can not be used as such to answer questions in open book examinations since questions are such that, they require to think and apply the information from books appropriately.

**4. More the number of books taken to the examination hall more will be the marks obtained**

This idea too is false. As not much time is available for a detailed study of books in the examination hall, examining / referring many book leads only to distraction of attention and waste of time. Hence, to select the right book, search and collect the required information quickly, only few important books are to be taken into the examination hall and used. This alone enables one to score good marks in the examination.

### 3:15:5 Advantages of Open Book Examination System

1. As this kind of examinations do not test student's memory, they need not cramp a lot of facts and figures.
2. Provides a chance for students to acquire knowledge during the preparation process of gathering suitable learning materials that they intend to take into the examination hall.
3. Develops the ability in students to find the informat on they require from books and other sources.
4. Enhances the comprehension and synthesizing skills in students as they need to reduce the content of books referred.

### 3:15:6 Disadvantages of Open-book Examinations

i) All students are not equally equipped, regarding the books they bring to the examination since only limited number of books will be available in the library on any topic. Further, even if students want to buy them for themselves, some books are costly.



ii) During examinations, more desk space is needed for each student to keep their books and refer them.

iii) Some students may spend too much of time in searching the relevant information for answering a question which results in that they could not answer most of the other questions.

iv) For most of the students, open-book examination is quite unfamiliar and hence they develop stress and anxiety.

v) It could not be expected that all students would have taken notes regarding the important concepts or ideas contained in the books they have read and carried to the examination hall.

vi) In open-book examinations most of the students do not score high marks.

## 3:16 Online Examinations

Online examination refers to the method of assessment for a remote candidate(s), which makes use of the internet connection and information technology process to find how much an individual, participating in the examination has knowledge related to a given topic or syllabus. In other words, 'online examination' means electronic assessment method in which a student, with the help of a computer having internet connection could view in the computer screen, the test-items contained in the question paper and record his/her answers for the same.

Those who wish to participate in the online examination should log in to the web site of the agency which conducts the examination, pay the examination fees in its account and register his/her name for the examination. The candidate taking the examination should know beforehand the date and time of the examination and sit before his/her own computer with internet connection or the one in a specific location like campus computer laboratory. The candidate should click the appropriate examination link and key in 'username' and 'password' in the computer. If one tries to enter into the particular website before the specified time, it will not get opened. On getting into the examination link, one can notice a clock on the right hand corner of the screen showing the time left. In the screen, pages of the question paper each containing six or seven questions will also appear. Candidates should choose and mark the pertinent answer by clicking with cursor. The candidate can notice on the bottom of the screen 'save' or 'submit' option. If save is clicked, the answer will be saved and such saved answer could be modified later on by going to that particular page. If submit button is clicked, then



the answers get submitted for assessment and no modification is possible thereafter. Moving the question paper page after page, the candidate should go on answering the questions. Then finally, the candidate should check whether the answers he/she has marked for all questions are right and if need be, effect necessary modifications and finally click 'submit' button. Answers are to be submitted within the given time. Once the stipulated time is over, the link will automatically gets closed. Online examination is an electronically managed assessment procedure. Generally, questions in the online examination will be in the form of multiple choice questions. Online examinations are used to evaluate cognitive abilities only.

Students' answers will be electronically evaluated, then and there and the examination results are usually notified in the computer screen immediately after they finish answering the examination. Examination results could also be received in the printed form. **'GMAT'** (Graduate Management Admission Test), **'GRE'** (Graduate Record Examination), **'SAT'** (Scholastic Assessment Test) are some examples for online examination.

### 3:16:1 Advantages of Online Examinations

i) There is no need for the students to take the examination only at a particular place and time.

ii) Nowadays online examinations are conducted all the twenty four hours on all days.

iii) Not much time is required to answer the questions. Time is required only for thinking and selecting the most appropriate among the answers given but to click the selected answer, a second or two is enough.

iv) Online examinations are user-friendly.

v) There is no scope for evil practices like 'copying' the answers from others.

vi) Students can take the examination at less cost.

### 3:16:2 Disadvantages of Online Examinations

1. Online examination is highly dependent on Internet connectivity; problems may arise in getting uninterrupted internet connectivity.

2. Answer in online assessment can only be right or wrong. There is no room for explaining an answer or getting partial credit.

3. There is no denying the fact that one can crack the online examination and get good marks merely by guesting and not by having solid knowledge.



4. This kind of examination is not suitable for essay type questions.

5. Without having basic functional knowledge in computer operations, students can not participate in online examinations.

## Conclusion

In this unit, important tools and techniques used for classroom assessment like observation, self-reporting techniques, anecdotal record, check list, rating scale, different kinds of tests like achievement test, diagnostic test, prognostic test, ability test, practical test etc. Rubrics and its meaning, important purposes of rubrics, steps in students learning to create a rubric for evaluating a classroom assignment, importance of a rubrics, affective domain and the important tools used to assess students' learning in it, particularly attitude scales (developed using the Thurstone method as well as Likert method), motivation scales and interest inventories (Kuder Preference Record, Strong Vocational Interest Blank, Thurstone Occupational Interest Inventory, Lee-Thorpe Occupation Interest Inventory), different types of test-items (Free response test-items like essay type questions, short answer type questions and paragraph questions, Controlled or Fixed response test-items otherwise known objective type questions like Multiple Choice, True or False, Matching the two columns, Completion type questions, Very short answer type questions, Finding the odd one from among those provided), their merits and demerits, and principles related to constructing different kinds of test-items, major issues in the assessment of students' learning viz. commercialization of assess-ment, poor test quality, domain dependency in evalua-tion, measurement issues and system issues, two major reforms in assessment ushered in viz. open book system of examinations and online examinations, have been discussed in detail.

**Review Questions:**

1. What is the difference between an assessment tool and an assessment technique? (C) [Ans: In 3:01, second paragraph only]
2. What is meant by 'observation'? State the important steps in carrying out a good observation. (B) [Ans: 3:02 + 3:02:1 + 3:02:2]
3. What is meant by 'observation'? Describe the different types of observation, their merits and demerits. State the important steps in carrying out a good observation. (A) [Ans: 3:02 + 3:02:3 + 3:02:3:01 to the end of 3:02:3:04 + 3:02:4 + 3:02:5 + 3:02:2]
4. Explain what is self-reporting technique. State its merits and demerits. (B) [Ans: 3:03 + 3:03:1 to the end of 3:03:3]



5. What do you mean by 'Anecdotal Record'? (C) [Ans: 3:04:1]
6. Give a definition for 'Anecdotal Record'. (C) [Ans: 3:04:2]
7. What is meant by an 'Anecdotal Record'? State its salient features. (B) [Ans: 3:04:1 + 3:04:3]
8. Explain the method of maintaining an anecdotal record. What are the uses of an anecdotal record? (B) [Ans: 3:04:4 + 3:04:5]
9. What are the limitations of anecdotal record? (C) [Ans: 3:04:6]
10. What is a 'Check list'? Mention its uses, merits and demerits. (B) [Ans: 3:05 + 3:05:1 to the end of 3:05:3]
11. What do you mean by a 'rating scale'? In what ways can a rating scale be constructed? Explain its merits and demerits. (A) [Ans: 3:06+ 3:06:1 + 3:06:2 + 3:06:2:01 to the end of 3:06:2:03 + 3:06:4 + 3:06:4:01 + 3:06:4:02]
12. Briefly describe different kinds of tests and their uses. (A) [Ans: 3:07:2 + those under (A), B & (C) of 3:07:2:01 + 3:07:2:02 and those under A, B and D of it + 3:07:2:03 and those under A, B and C + 3:07:2:04 and those under A and B of it]
13. Explain with an illustration the concept of a 'rubric' and discuss its purpose and importance. (A) [Ans: 3:08 + 3:08:2 + 3:08:1 + 3:08:4]

14. State the steps in students learning to create a rubric. (C) [Ans: 3:08:3]

15. Explain the important behaviour elements learned in the affective domain and mention the tools used to assess them. (B) [Ans: 3:09 + 3:09:1 + 3:9:2]

16. Explaining what are attitude scales, discuss any one of the methods of developing an attitude scale. (A) [Ans: 3:09:3 + 3:09:3:02]

17. What do you mean by 'attitude'? Explain the Thurstone attitude scale. Mention the uses and limitations of attitude scales. (A) [Ans: 3:09:3 + 3:09:3:01 + 3:09:3:03 + 3:09:3:04]

18. What is meant by 'Motivation'? Describe a motivation scale. (A) [Ans: 3:09:4:01 + 3:09:4:02 + In 3:09:4:03, those under 2 only]

19. What is meant by 'Interest'? Describe any one of the interest inventories. (A) [Ans: 3:10:1 + 3:10:2 + 3:10:3 + 3:10:4:02]

20. Write a short note on the following :
    i) Kuder Preference Record (A) [Ans: 3:10:4:01]
    ii) Strong-Campbell Interest Inventory (B) [Ans: 3:10:4:03]
    iii) Thurstone Occupational Interest Schedule (B) [Ans: 3:10:4:04]
    iv) Lee - Thorpe Occupation Interest Inventory (B) [Ans: 3:10:4:05]



    v) Likert Summative Rating Scale (B) [Ans: 3:09:3:02]

21. Explain the various types of test items, with examples (A) [Ans. 3:11 + 3:11:1 + 3:11:1:01 to the end of 3:11:1:05]

22. State the merits and demerits of objective type questions. (B) [Ans. 3:11:2 + 3:11:3]

23. Write a note on 'Short answer type questions'. (C) [Ans. 3:11:4 + 3:11:4:01]

24. Write a short note on the following :
    i) 'Paragraph type' question (C) [Ans: 3:11:5]
    ii) 'Essay type' question and its merits and demerits. (B) [Ans: 3:11:6 + 3:11:6:01 + 3:11:6:02]

25. Explain the principles involved in constructing various forms of test items. (A) [Ans. 3:12 + 3:12:1 to the end of 3:12:6]

26. Discuss any three major issues in the assessment of student learning. (A) [Ans: 3:13 + From 3:13:1 to the end of 3:13:3]

27. Explain the important measurement issues in assessing learning achievement. (A) [Ans: 3:13:4 + 3:13:4:01 to the end of 3:13:4:04]

28. Discuss the problems related to evaluation system. (B) [Ans: 3:13:5 + 3:13:5:01 to the end of 3:13:5:03]